

ASUS ZC600KL

RF Hardware Trouble Shooting Guide

ASUS[®] Inspiring Innovation • Persistent Perfection

Inspiring Innovation. Persistent Perfection

Outline

(*) Debug Tool Required

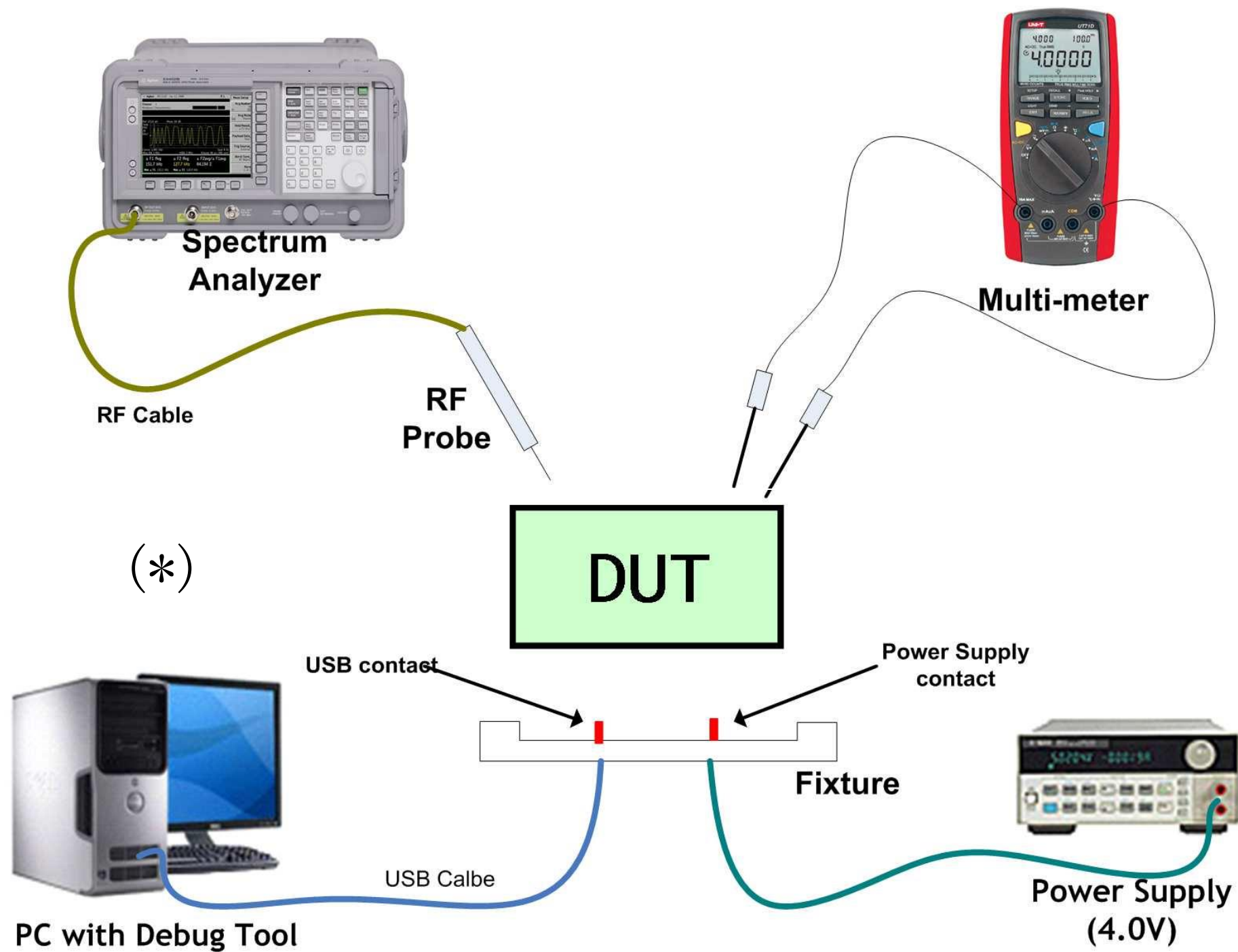
- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/Band8/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

Outline

(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band8/Band6/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

Repair and Test Environment

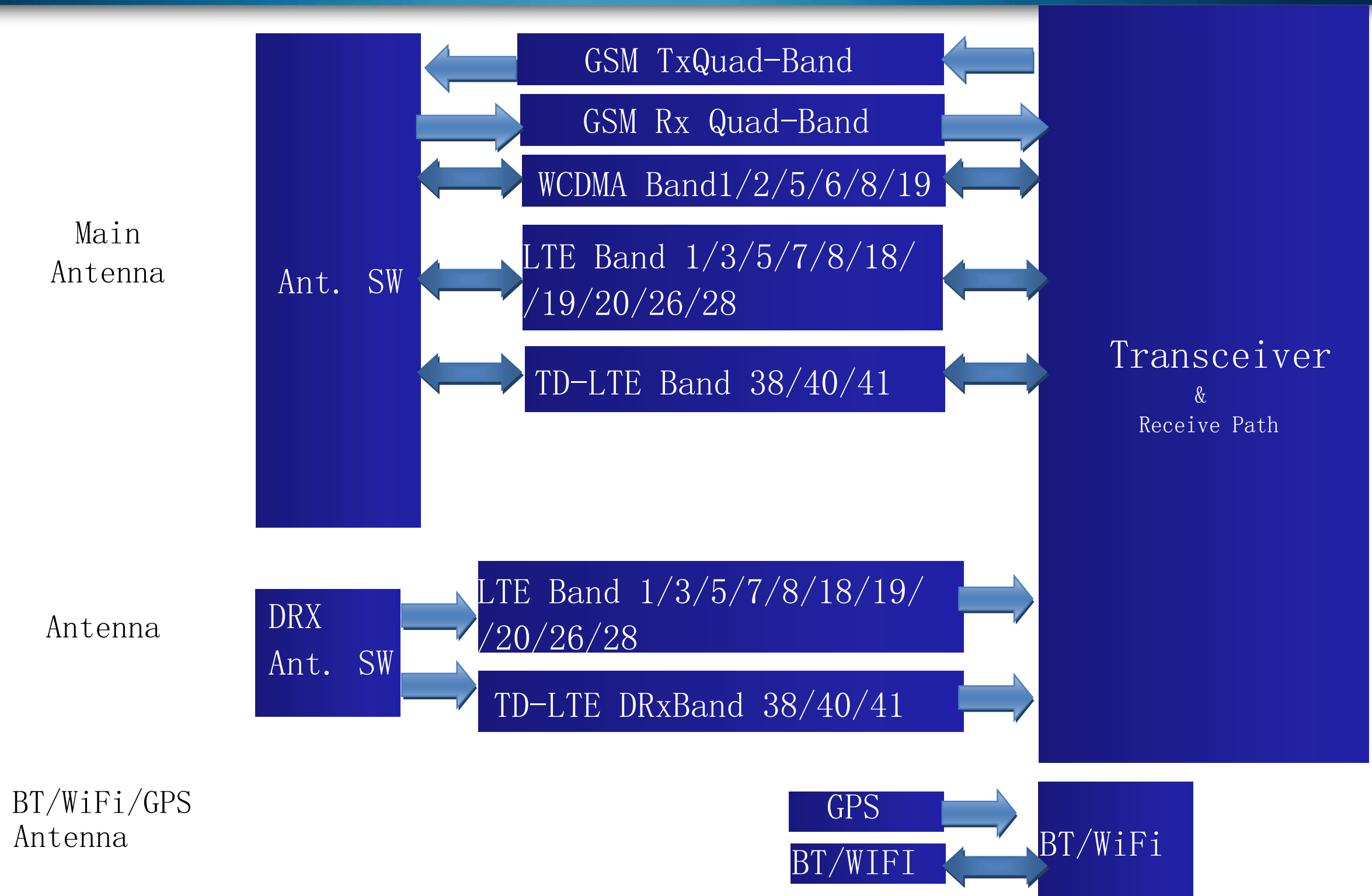


Outline

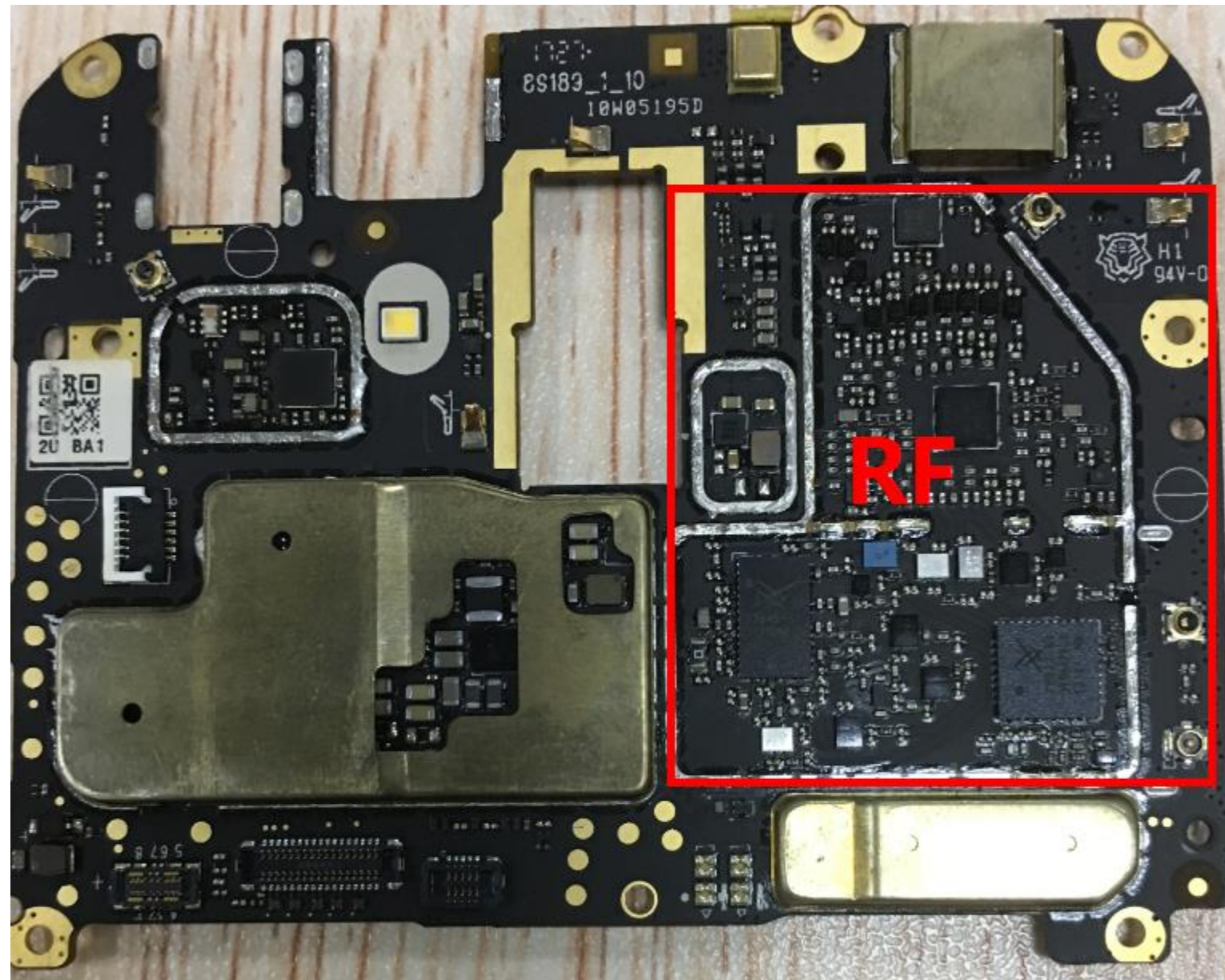
(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/Band8/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

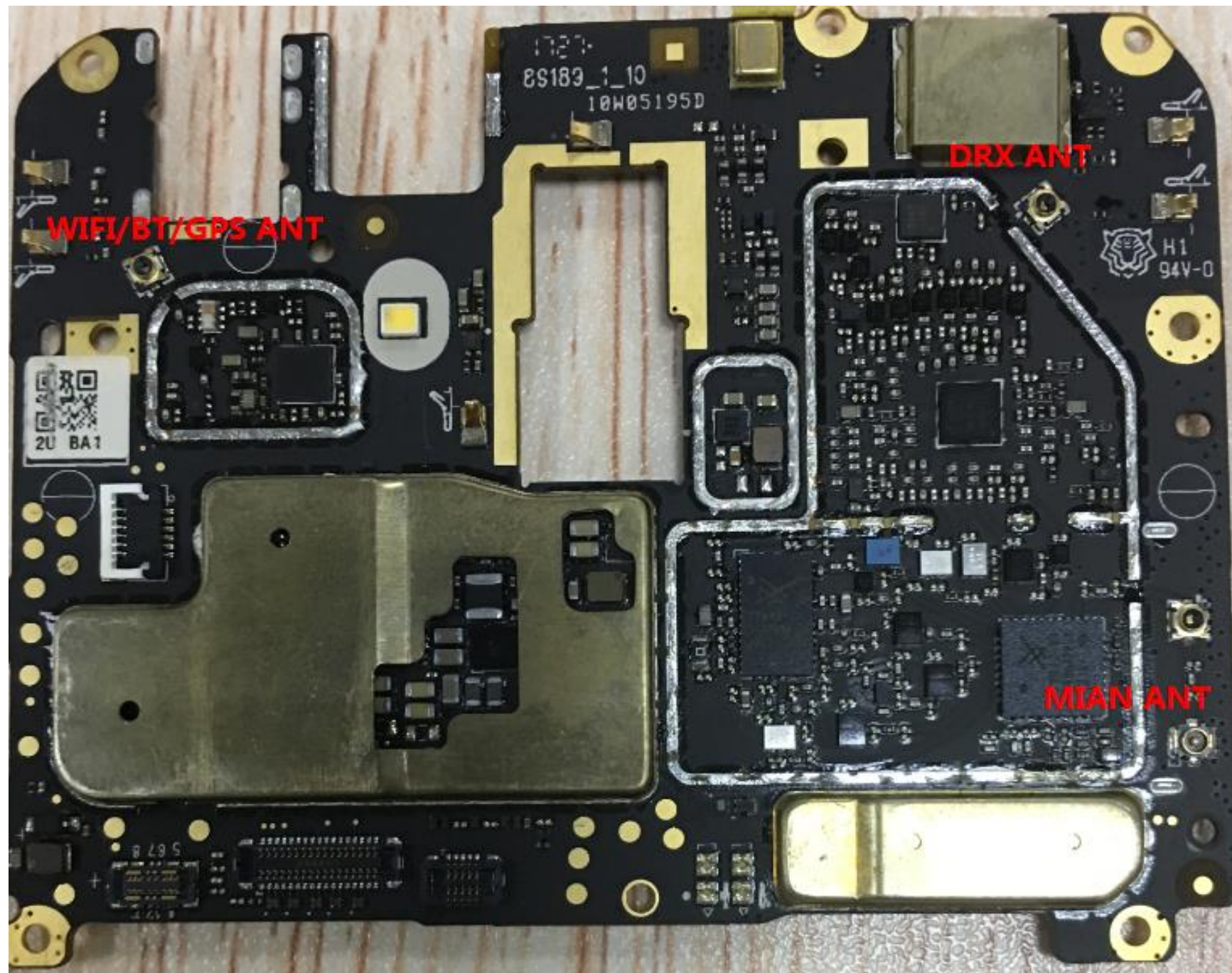
RF Block Diagram



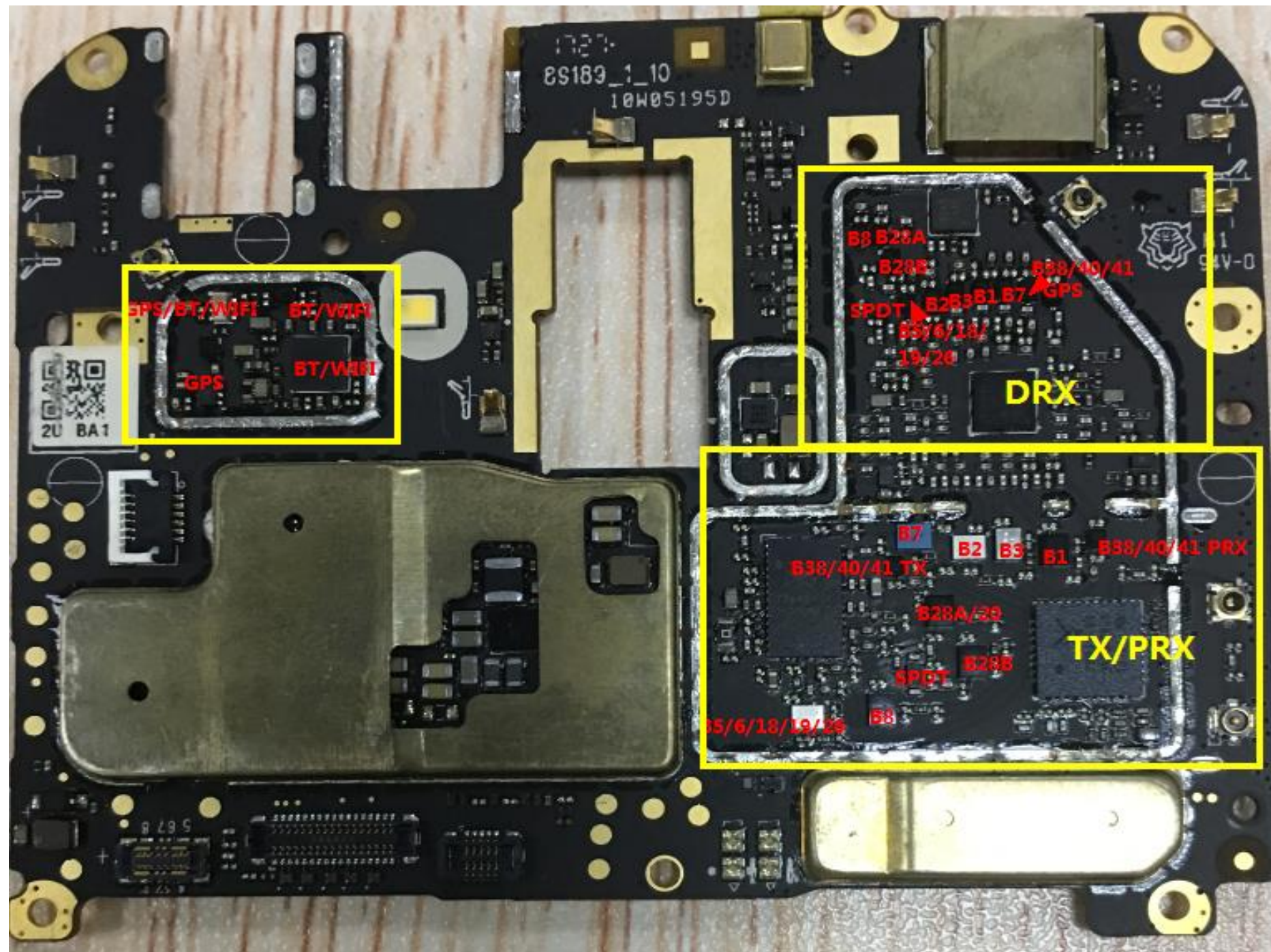
ZC600KL RF Areas Top Side



ZC600KL RF Areas Bottom Side



Key components locations (Bot)



Band Switch

Ant. Switch

PA

Duplexer

WCN

SAW

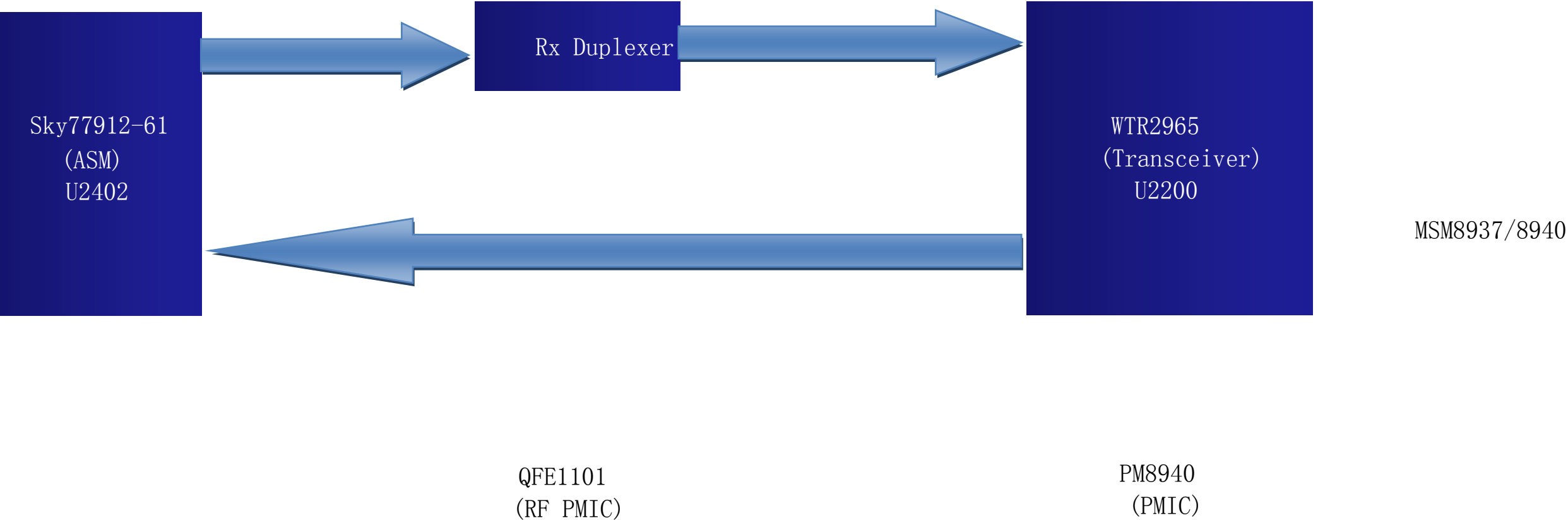
Outline

(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/BAND8/BAND19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE Div. Rx 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

GSM Block Diagram

Main Ant.



2G TX and EGSM/GSM850 RX

Tx path

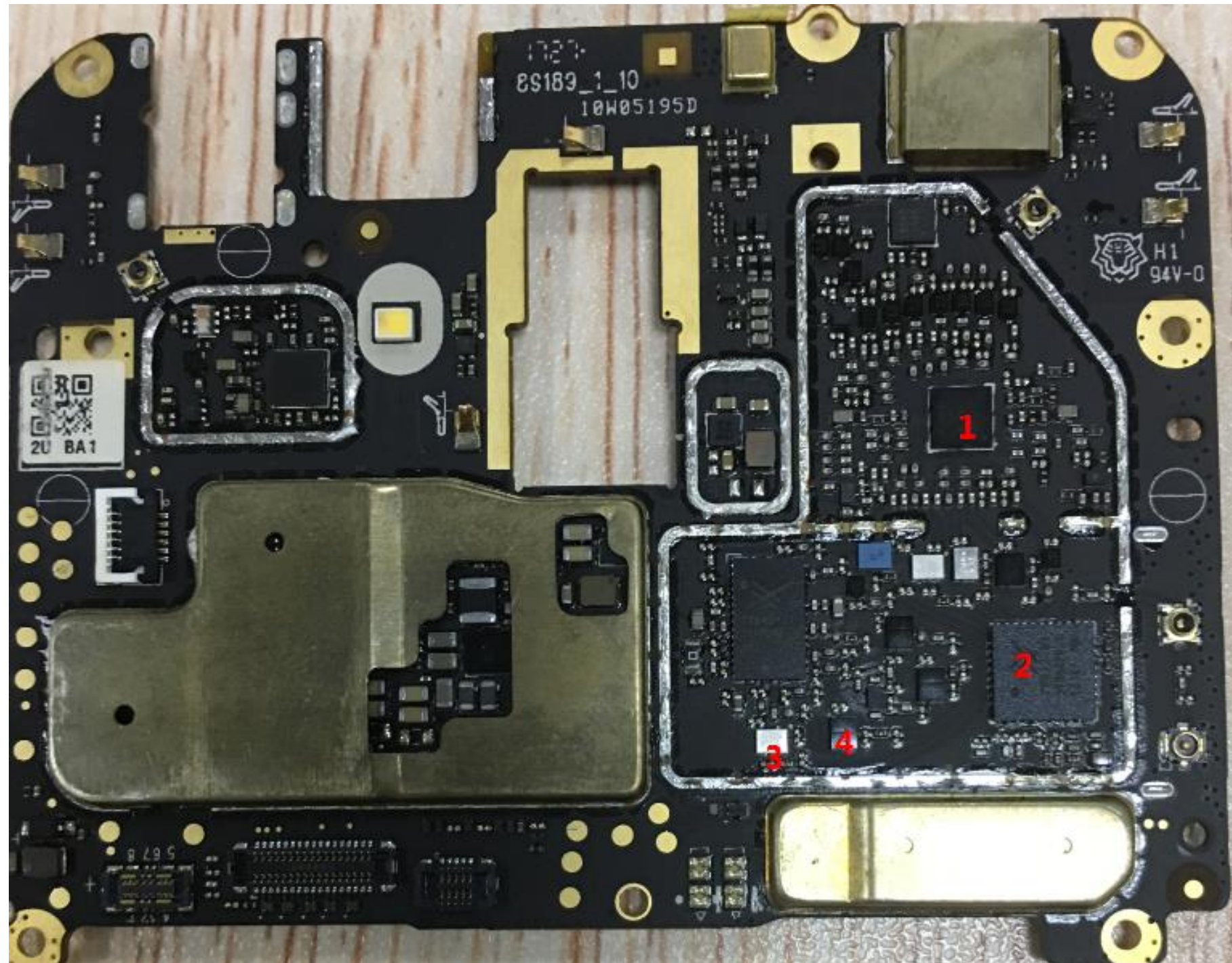
Rx path

1. U2200: Transceiver
[WTR2965](#)
2. U2402 : TX Module
[Sky77912-61](#)
3. U2505 : B5 Duplexer
4. U2508: B8 Duplexer

Serial IC

Tx comp.

Rx comp.



DCS and PCSRx

Tx path

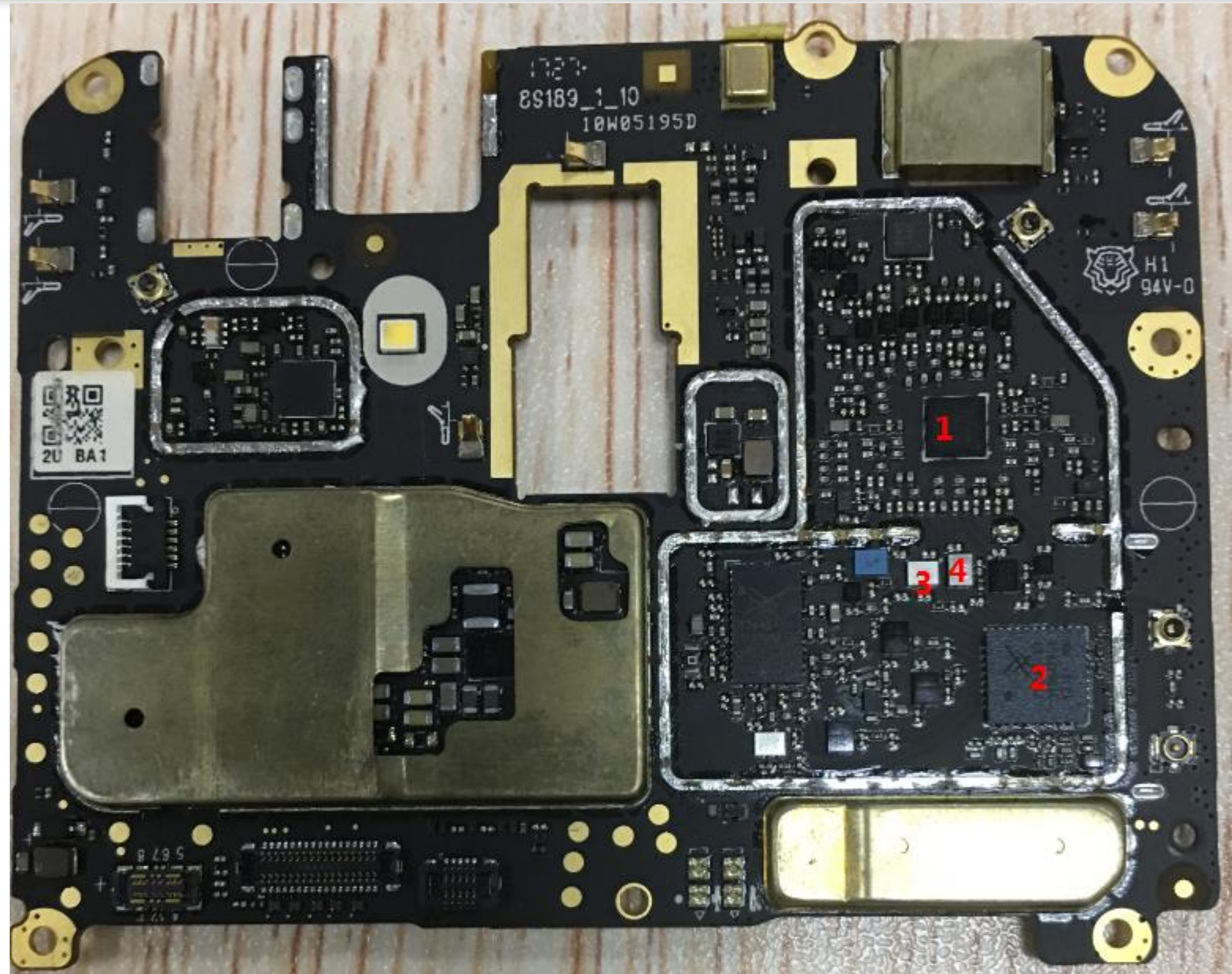
Rx path

1. U2200: Transceiver
[WTR2965](#)
2. U2402 : TX Module
[Sky77912-61](#)
3. U2502 : B2 Duplexer
4. U2503 : B3 Duplexer

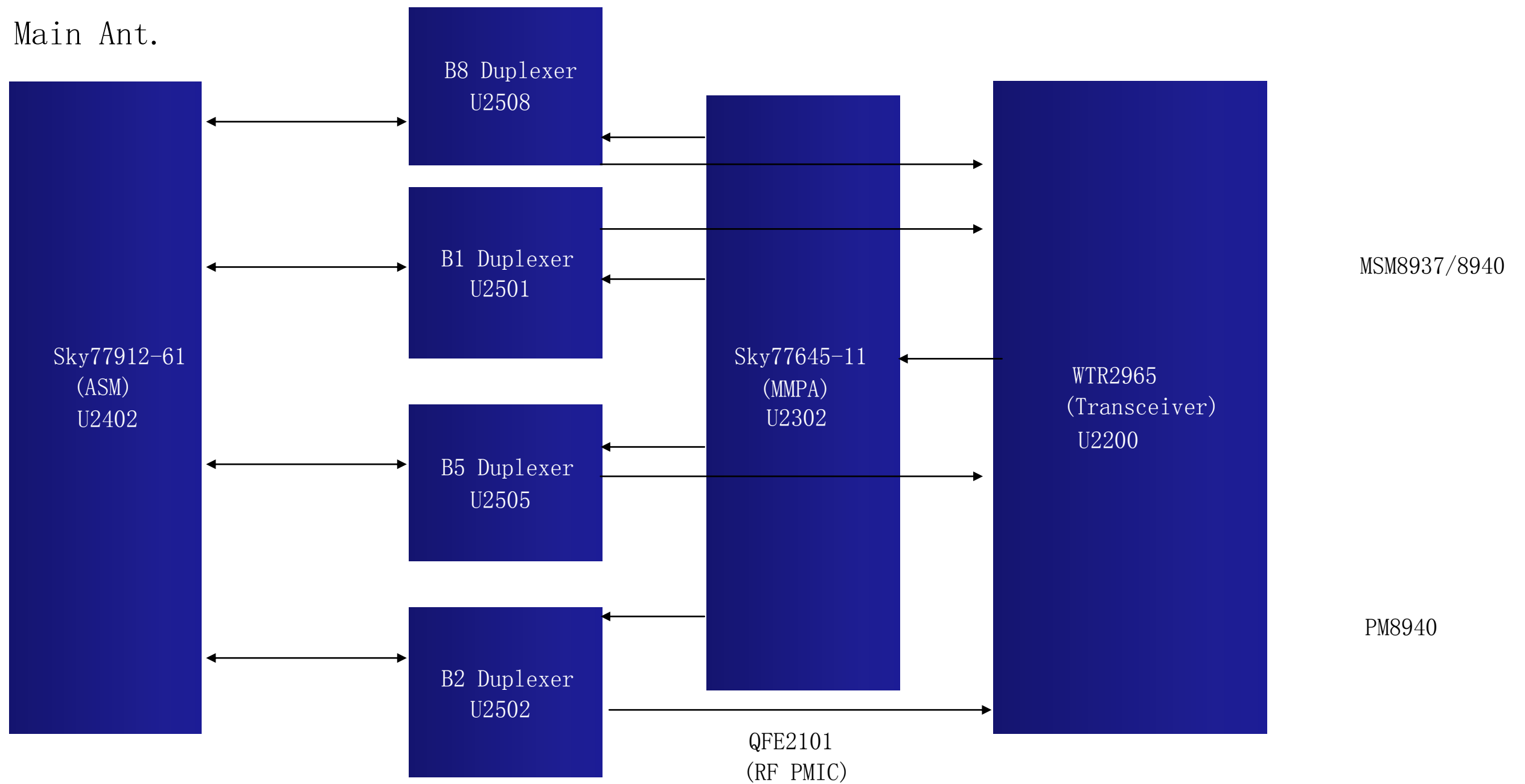
Serial IC

Tx comp.

Rx comp.



WCDMA Main Ant Block Diagram



WCDMA Band1

Tx path

Rx path

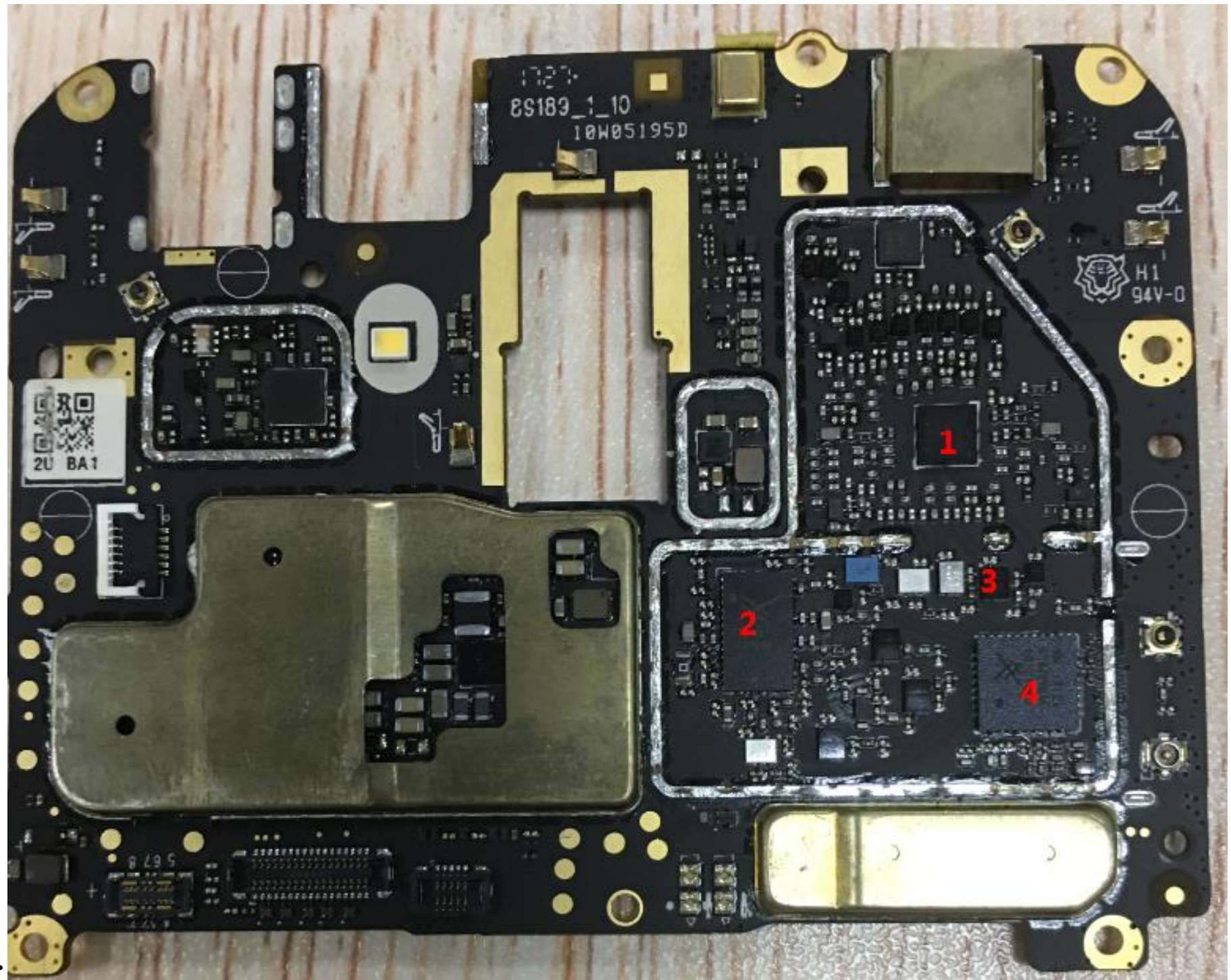
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2501 : B1 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



WCDMA Band2 (TW)

Tx path

Rx path

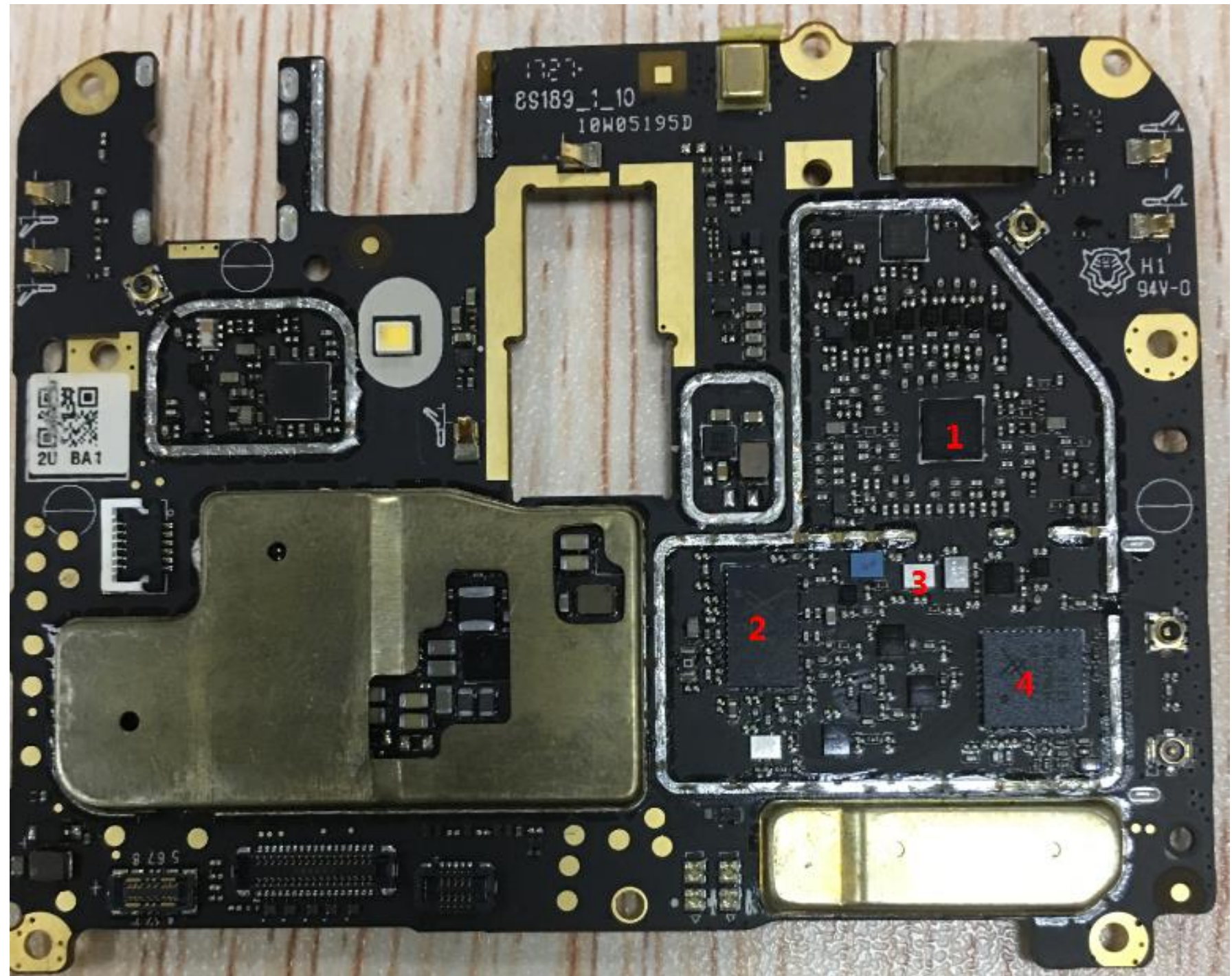
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2502 : B2 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



WCDMA Band5/6/19

Tx path

Rx path

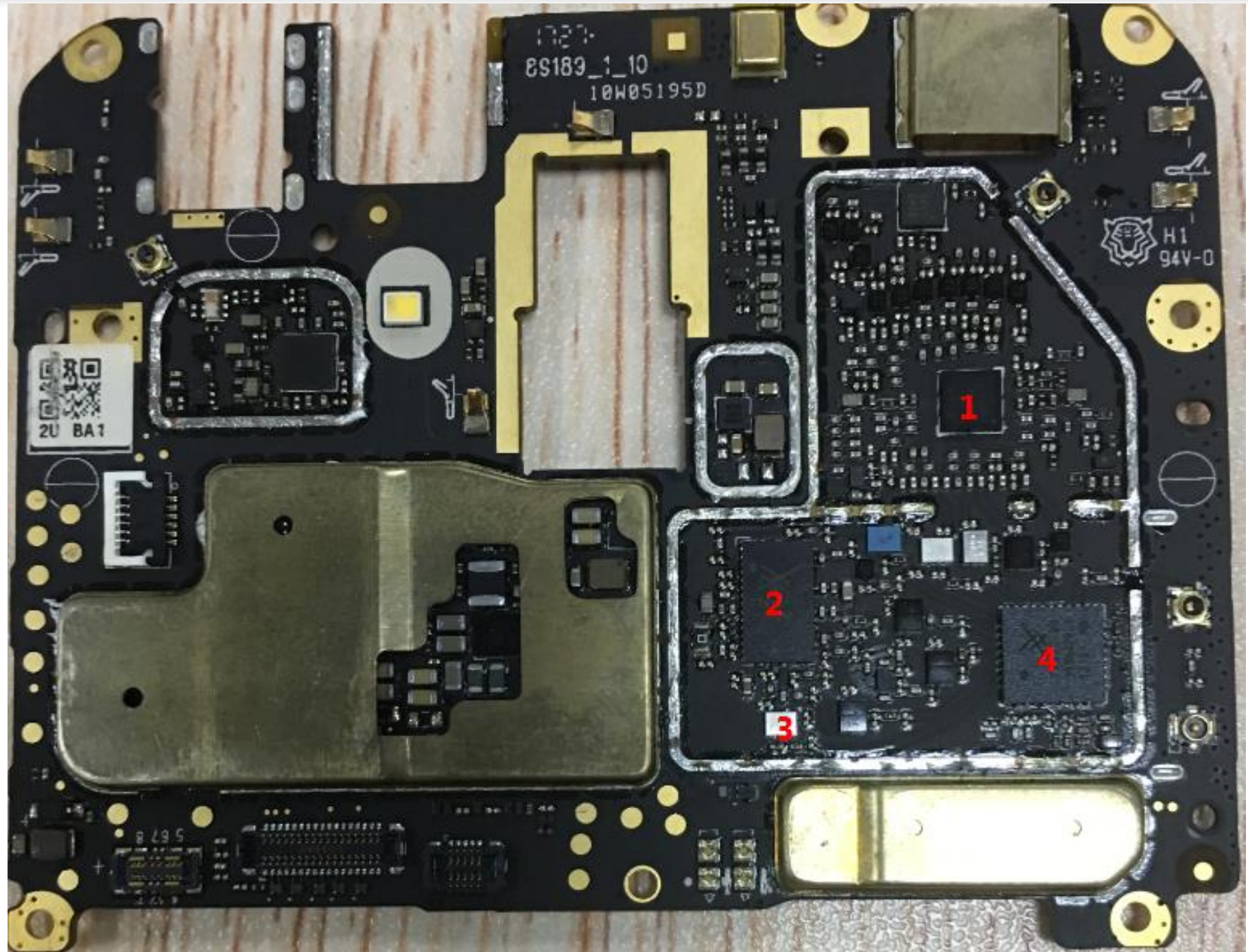
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2505 : B5 Duplexer
4. U2402 : TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



WCDMA Band8

Tx path

Rx path

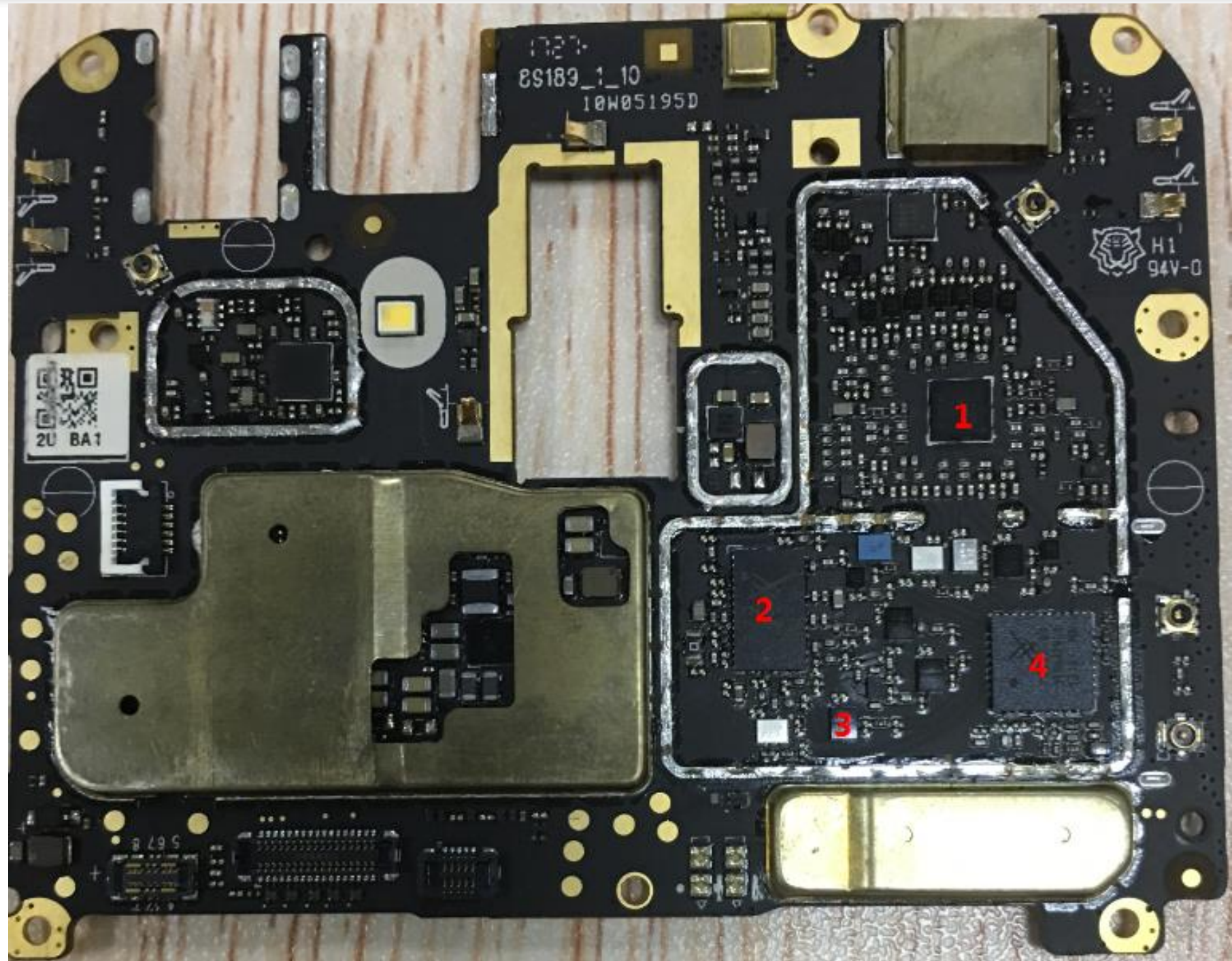
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2508 : B8 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

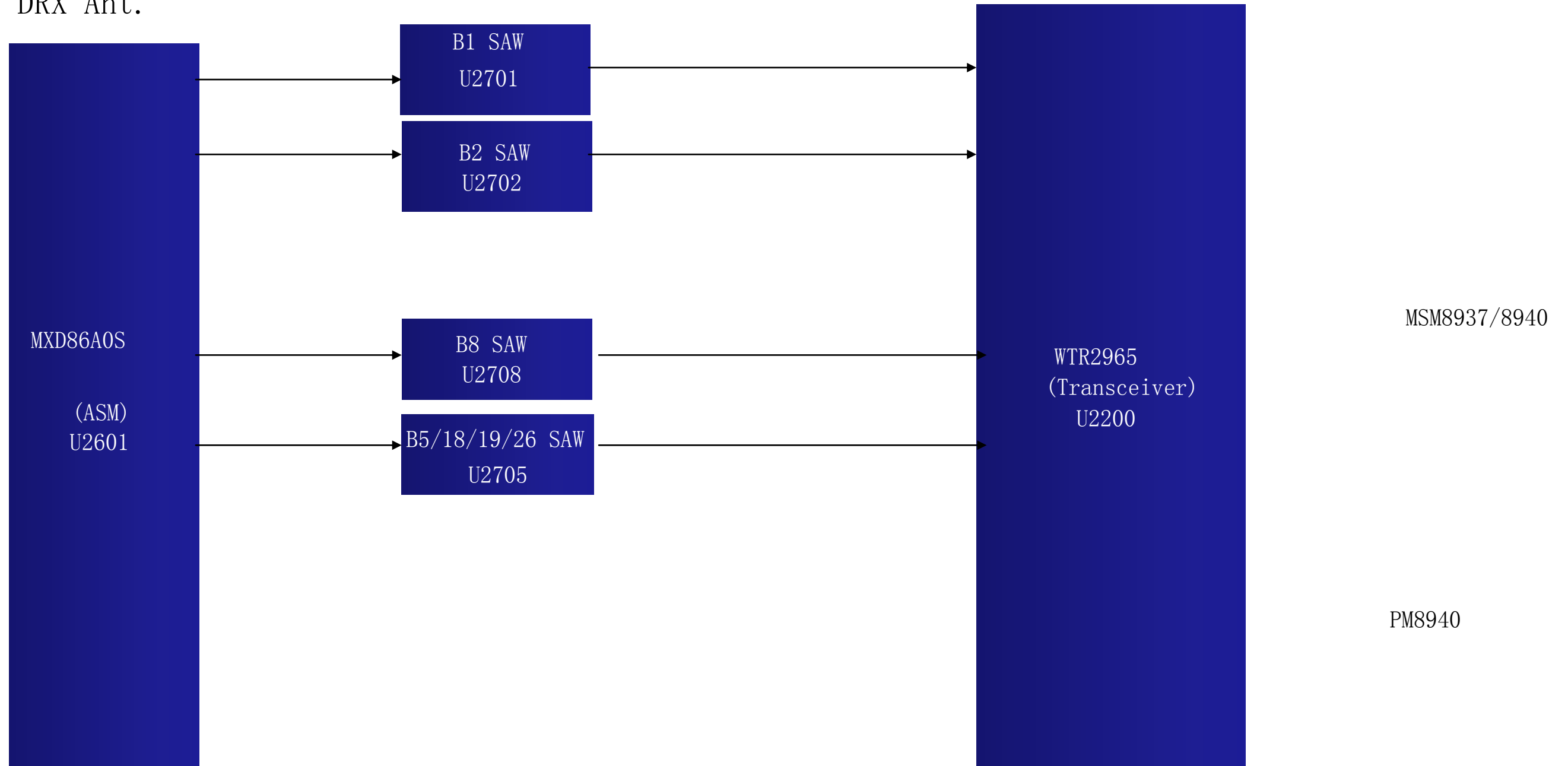
Rx comp.

Tx/Rx comp.



WCDMA Div. Rx Block Diagram

DRX Ant.



WCDMA Band 1

Tx path

Rx path

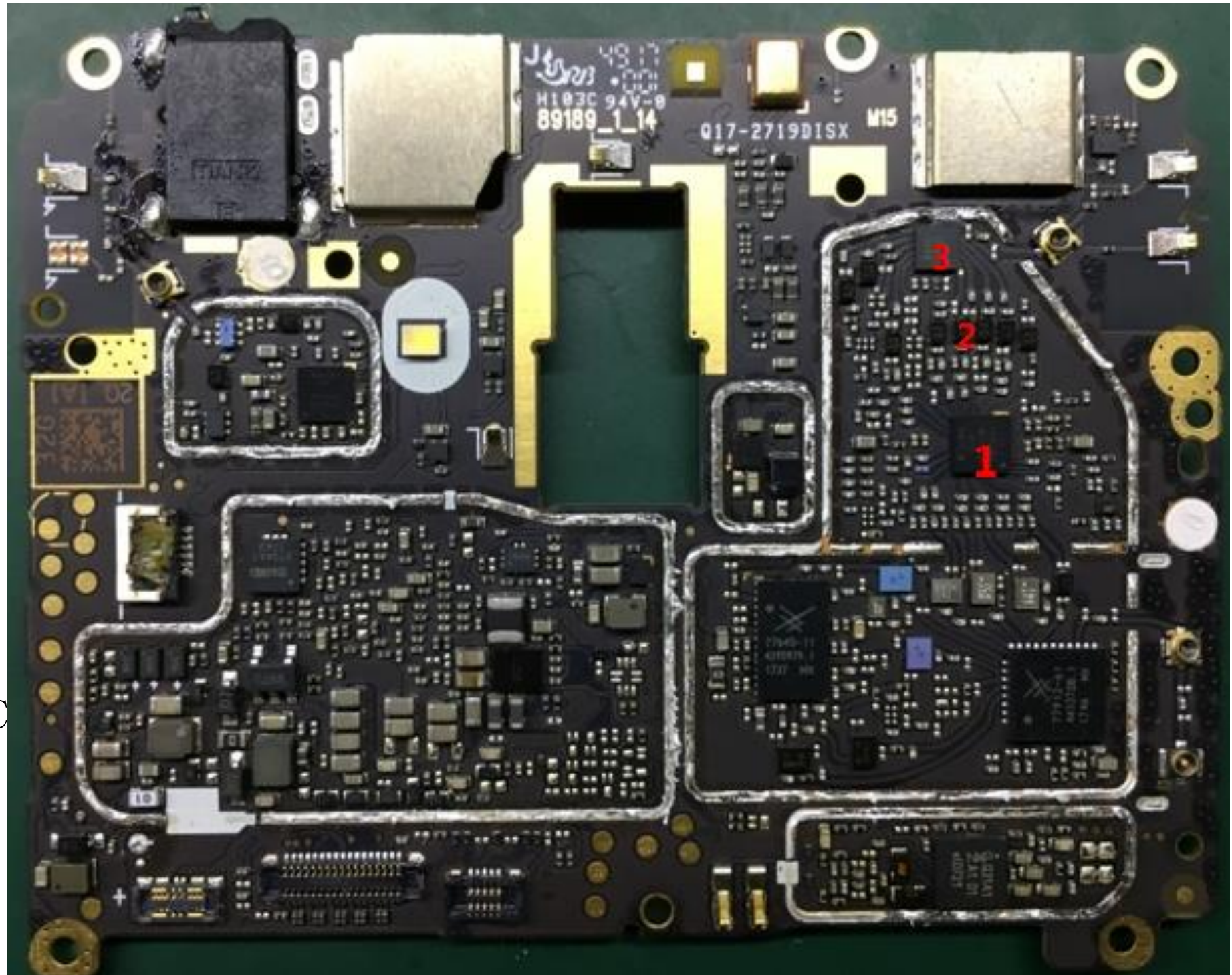
1. U2200: Transceiver
[WTR2965](#)

2. U2701 : B1 SAW

3. U2601: Ant. Switch
[MXD86A0S](#)

Serial IC

Rx comp.



WCDMA Band 2 (TW)

Tx path

Rx path

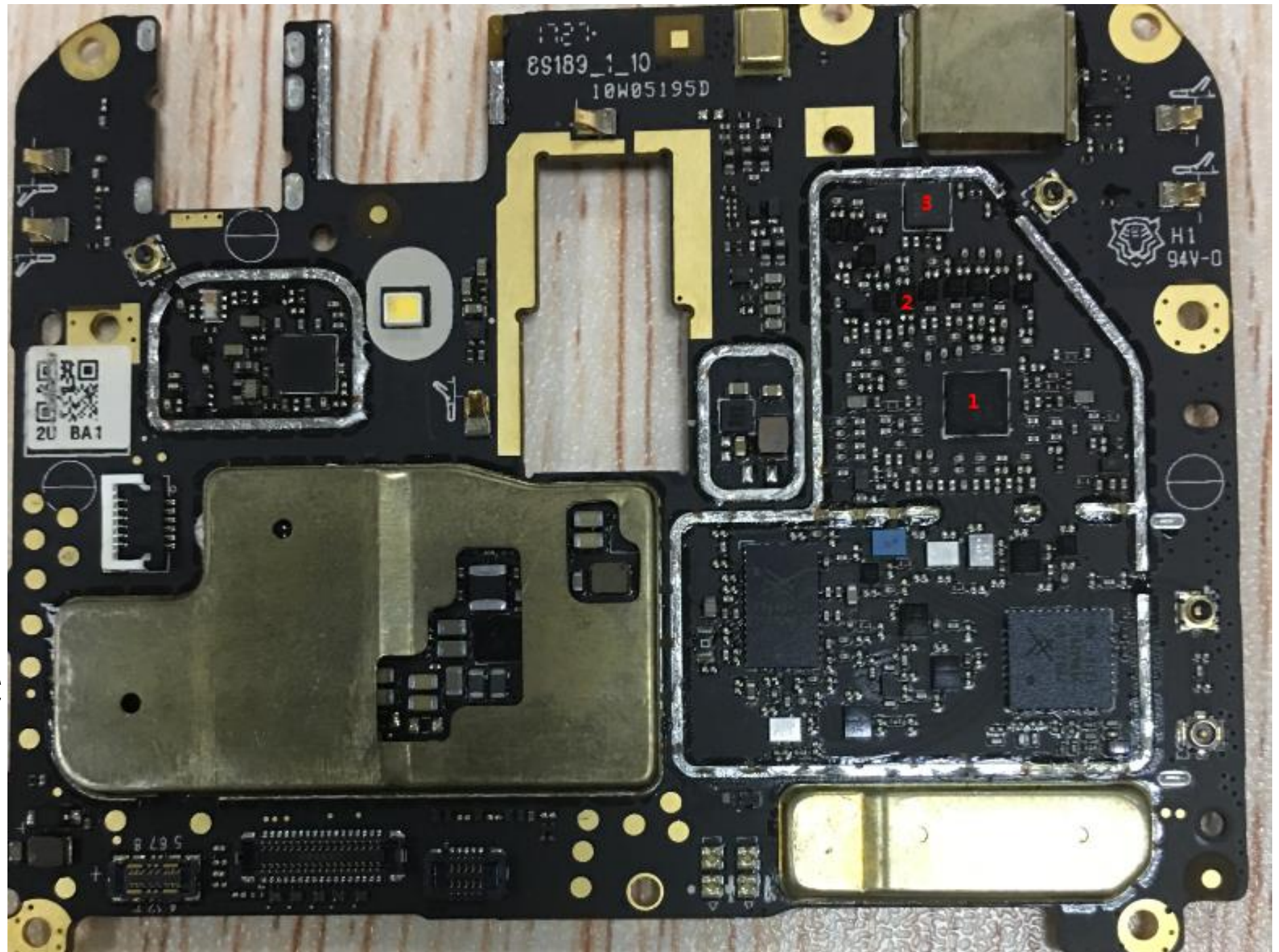
1. U2200: Transceiver
[WTR2965](#)

2. U2702 : B2 SAW

3. U2601: Ant. Switch
[MXD86A0S](#)

Serial IC

Rx comp.



WCDMA Band 5/6/19

Tx path

Rx path

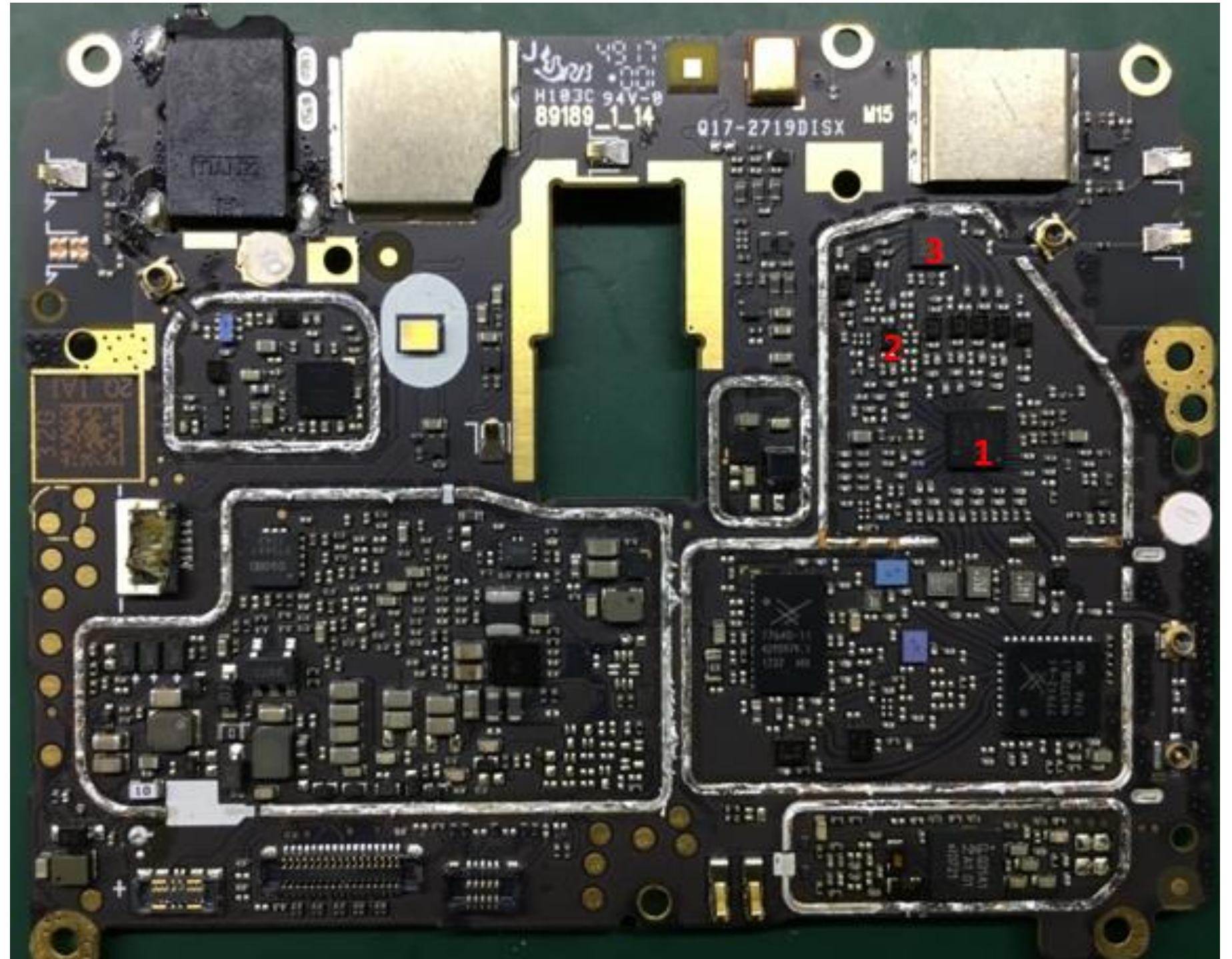
1. U2200: Transceiver
[WTR2965](#)

2. U2702 : B5/6/19 SAW

3. U2601: Ant. Switch
[MXD86A0S](#)

Serial IC

Rx comp.



WCDMA Band 8

Tx path

Rx path

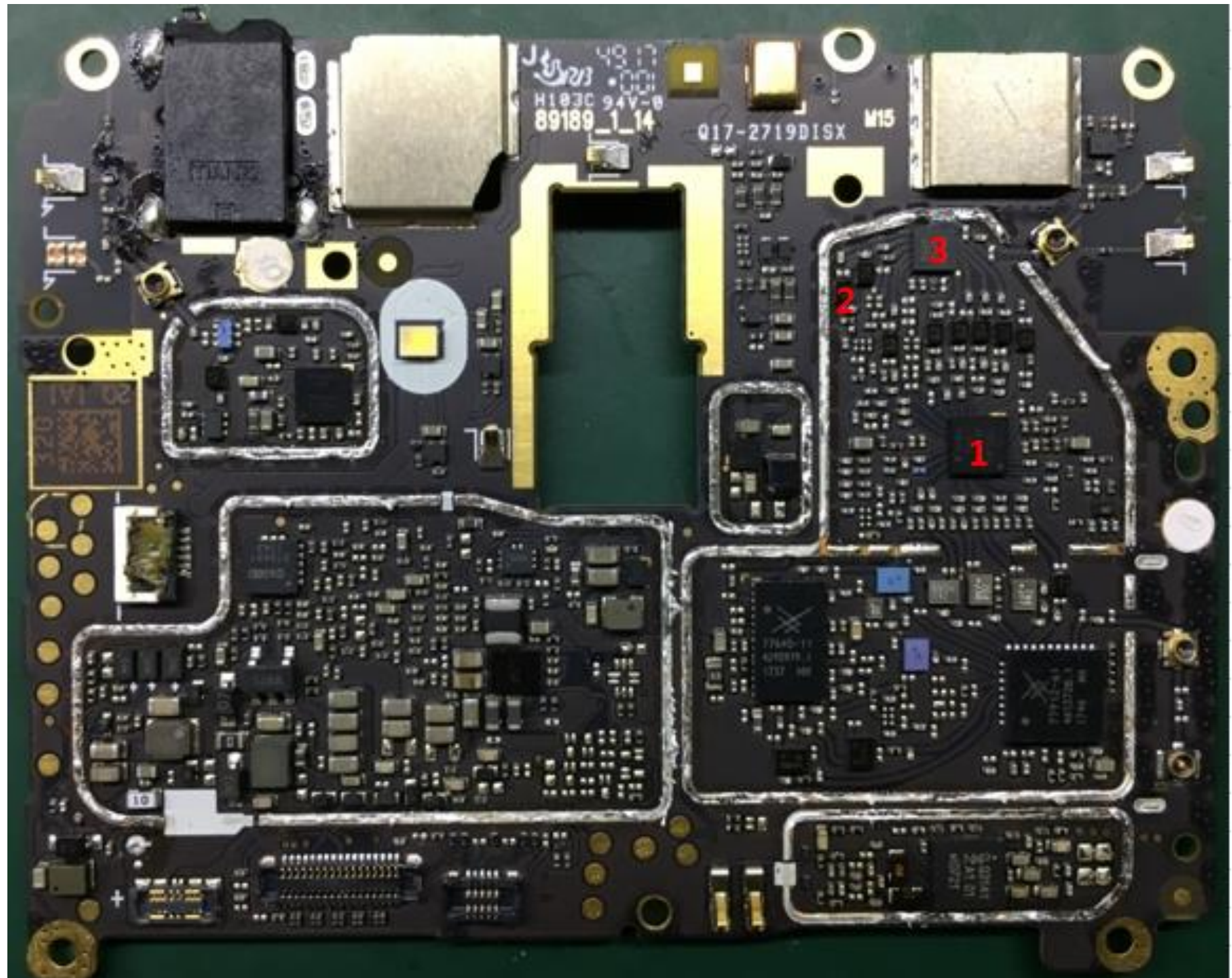
1. U2200: Transceiver
[WTR2965](#)

2. U2702 : B8 SAW

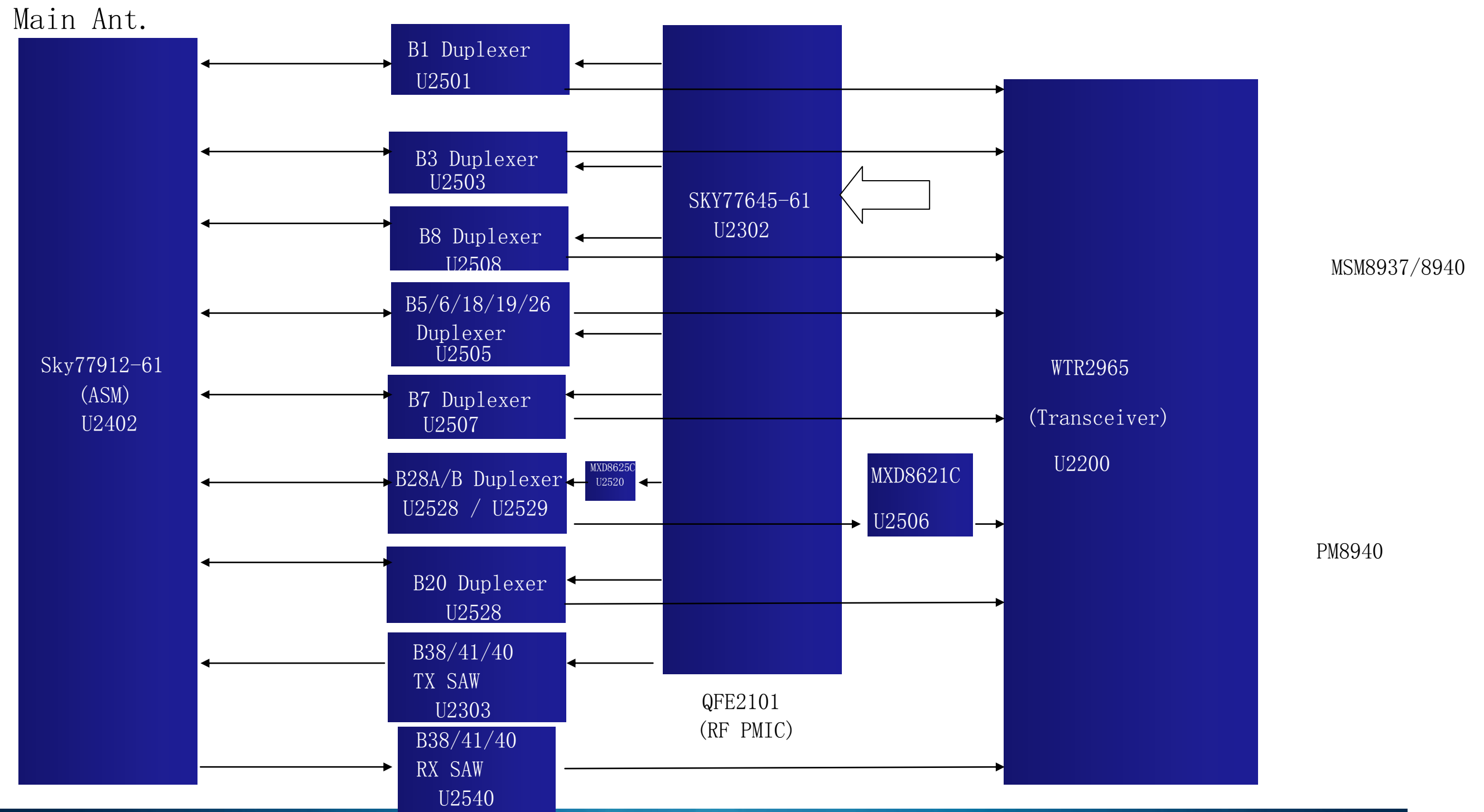
3. U2601: Ant. Switch
[MXD86A0S](#)

Serial IC

Rx comp.



LTE Main Ant Block Diagram



LTE Band 1

Tx path

Rx path

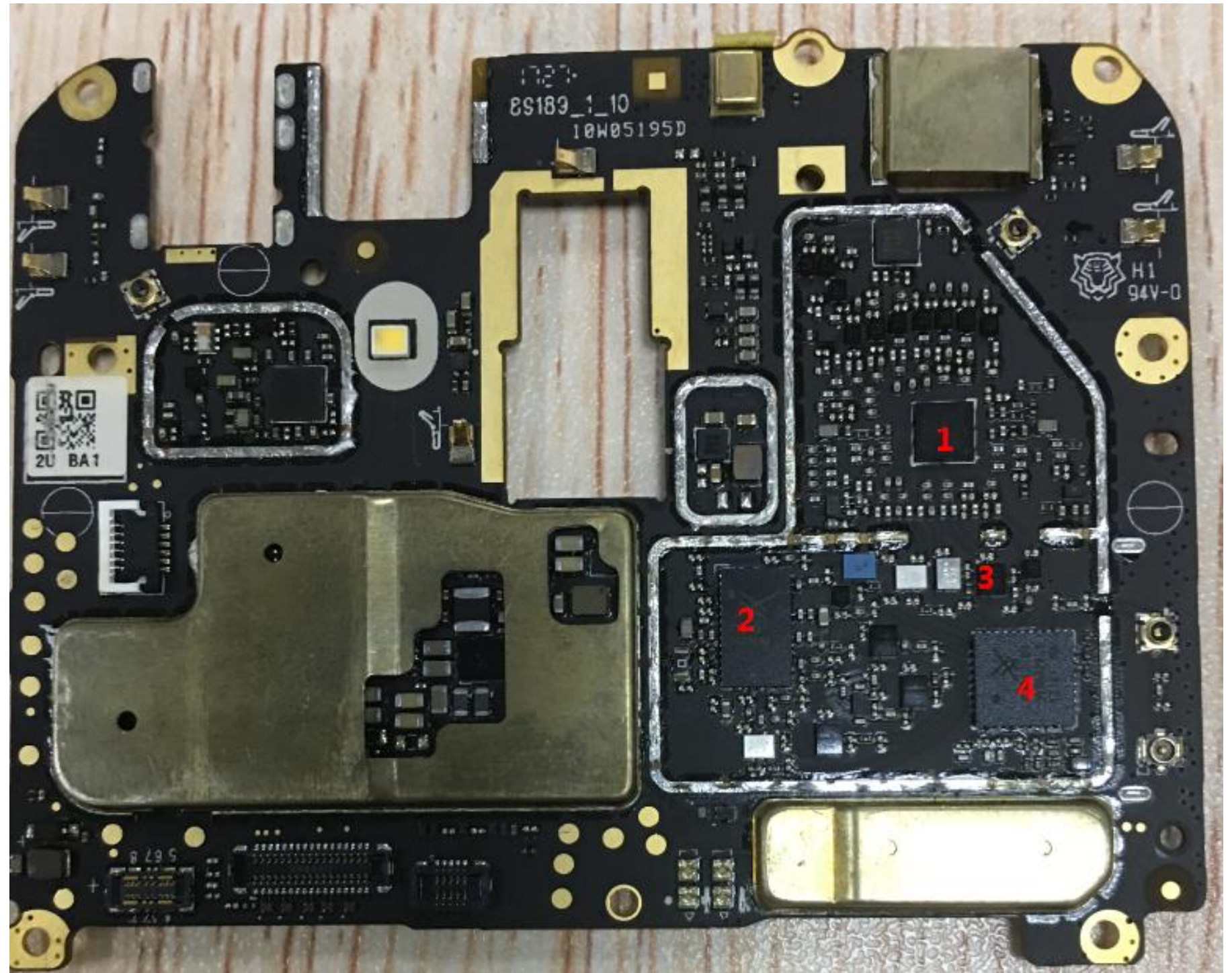
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2501 : B1 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 3

Tx path

Rx path

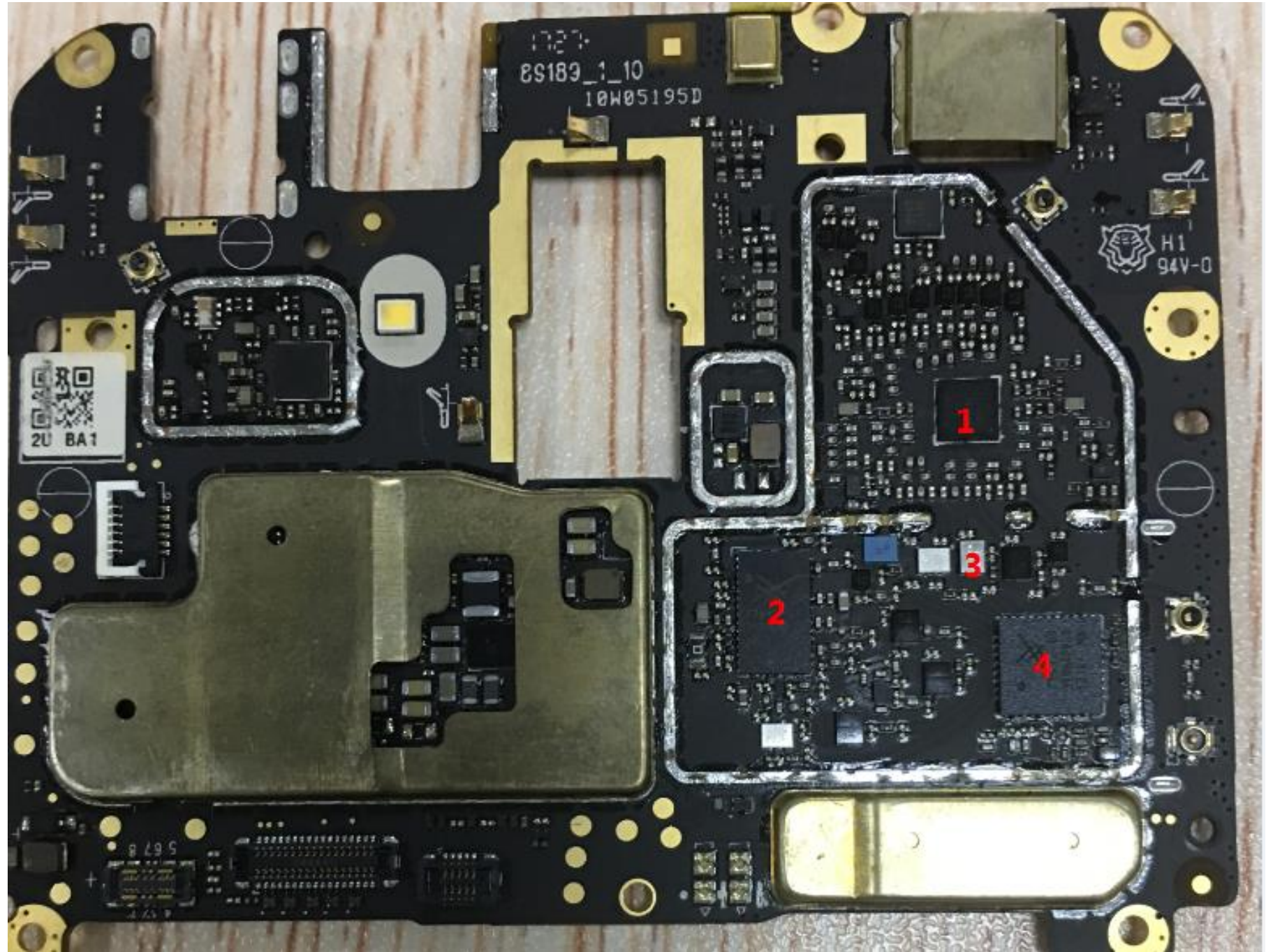
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2503 : B3 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 5/18/19/26

Tx path

Rx path

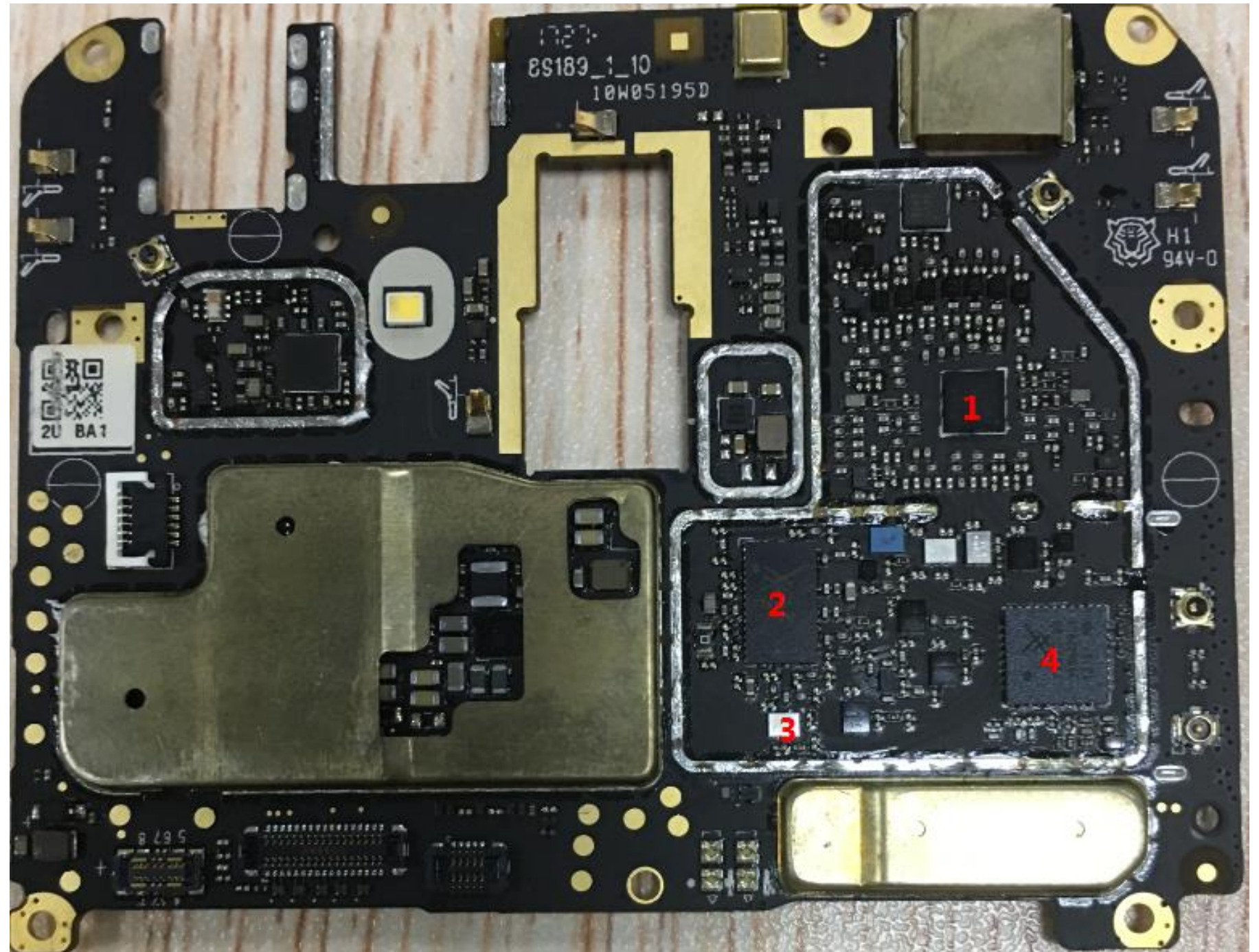
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2505 : B5 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 7

Tx path

Rx path

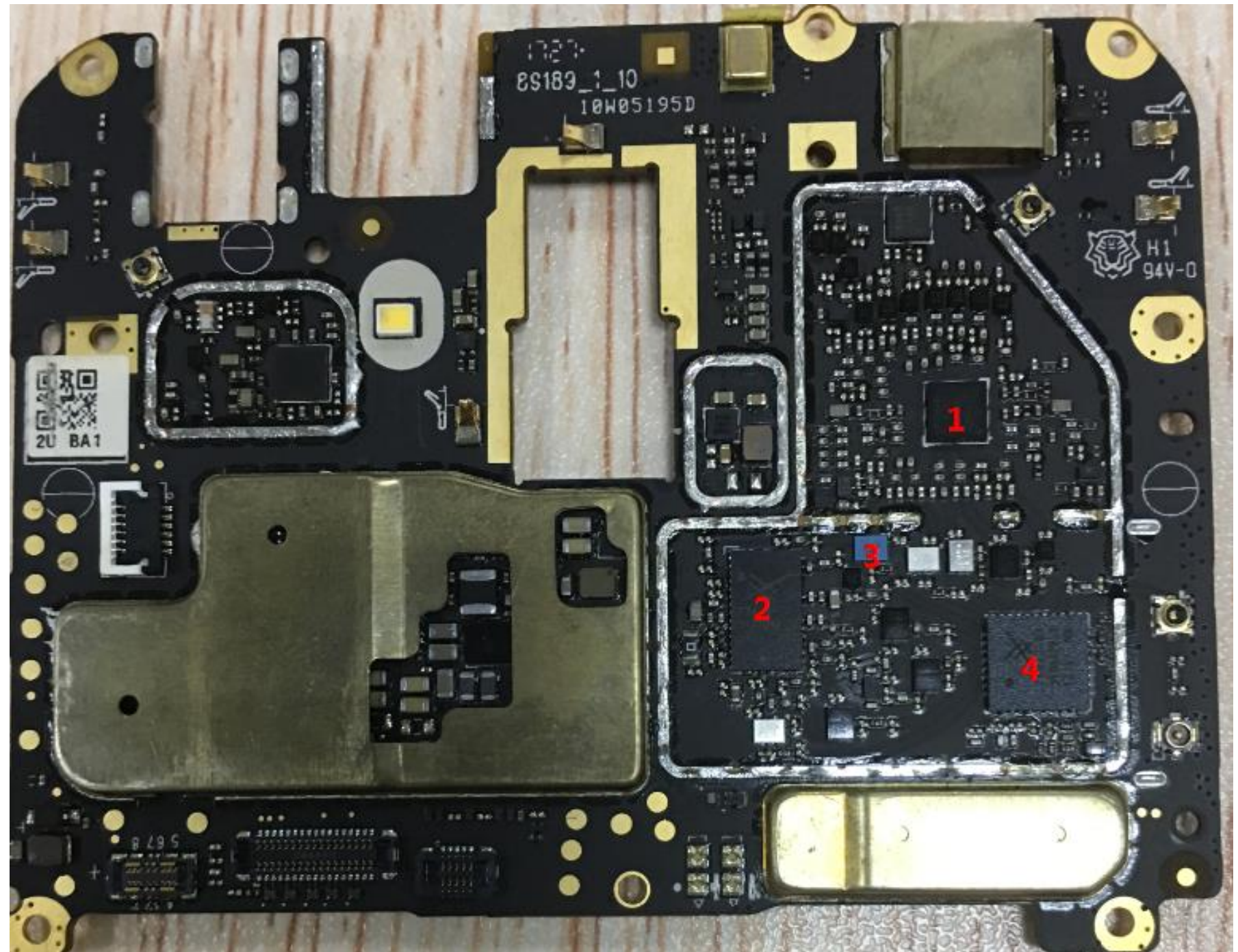
1. U2200: Transceiver
WTR2965
2. U2302 : Power Amp.
Sky77645-11
3. U2507 : B7 Duplexer
4. U2402: TX Module
Sky77912-61

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 8

Tx path

Rx path

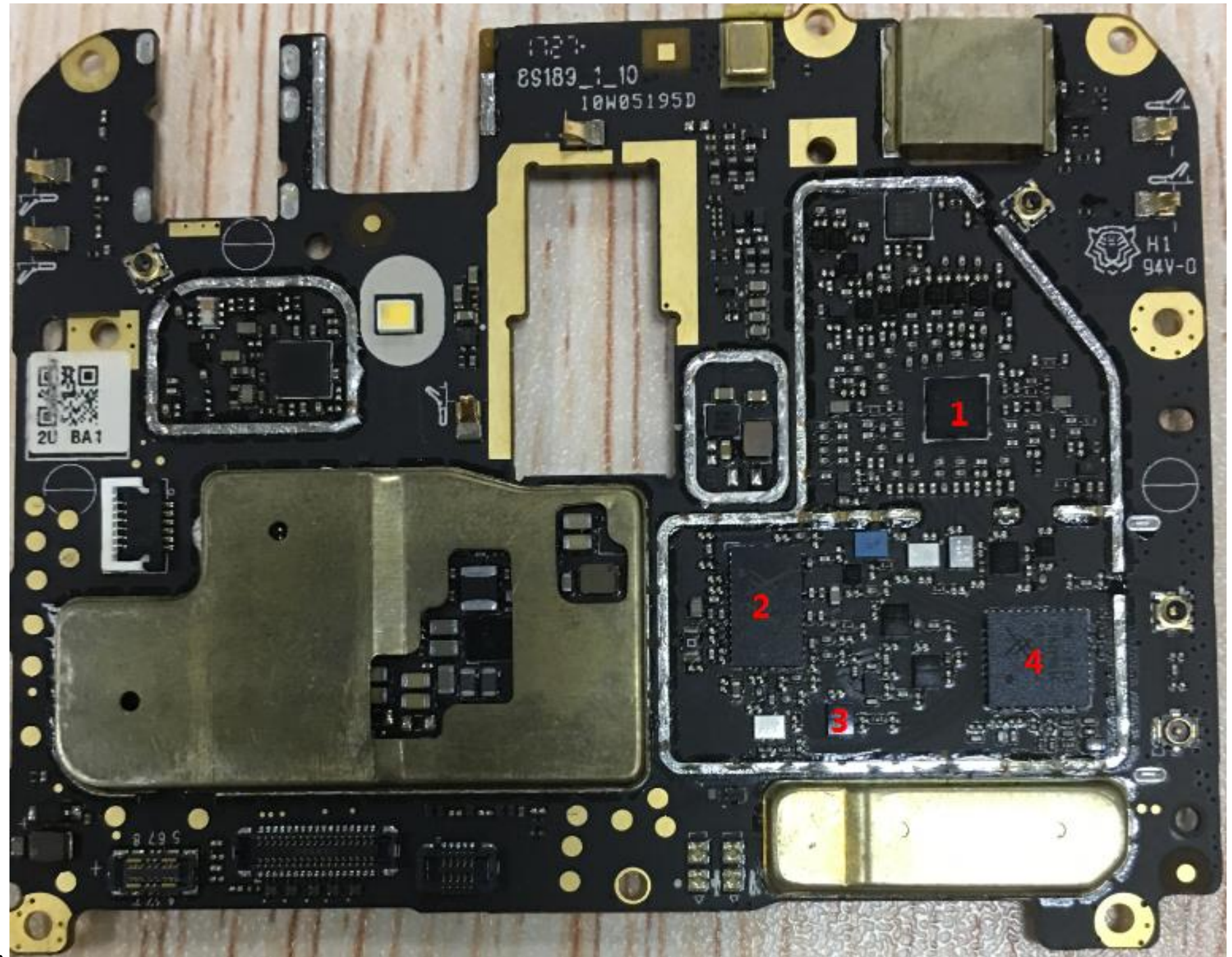
1. U2200: Transceiver
WTR2965
2. U2302 : Power Amp.
Sky77645-11
3. U2508 : B8 Duplexer
4. U2402: TX Module
Sky77912-61

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 20 (WW)

Tx path

Rx path

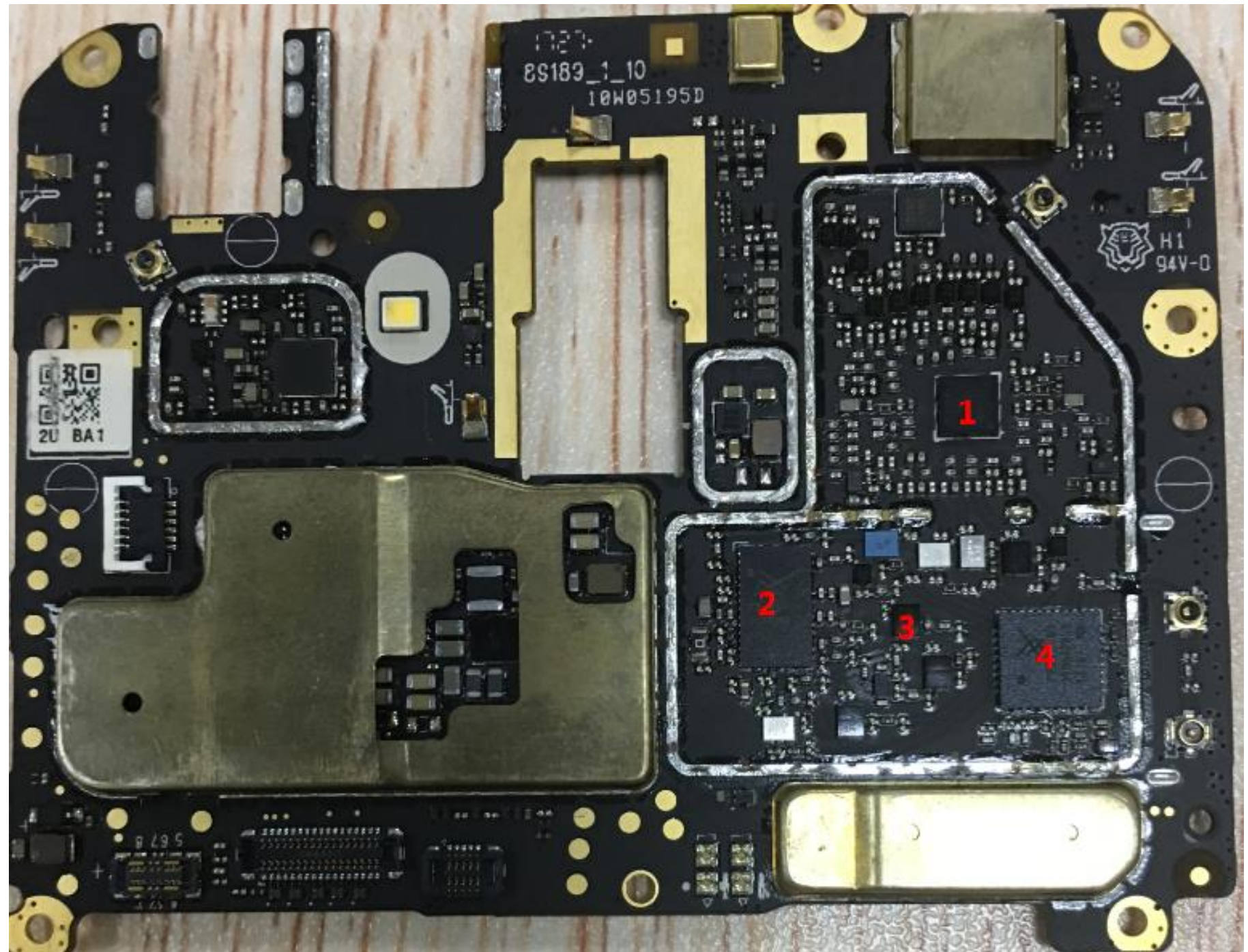
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2528 : B20 Duplexer
4. U2402: TX Module
[Sky77912-61](#)

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



LTE Band 28 (TW)

Tx path

Rx path

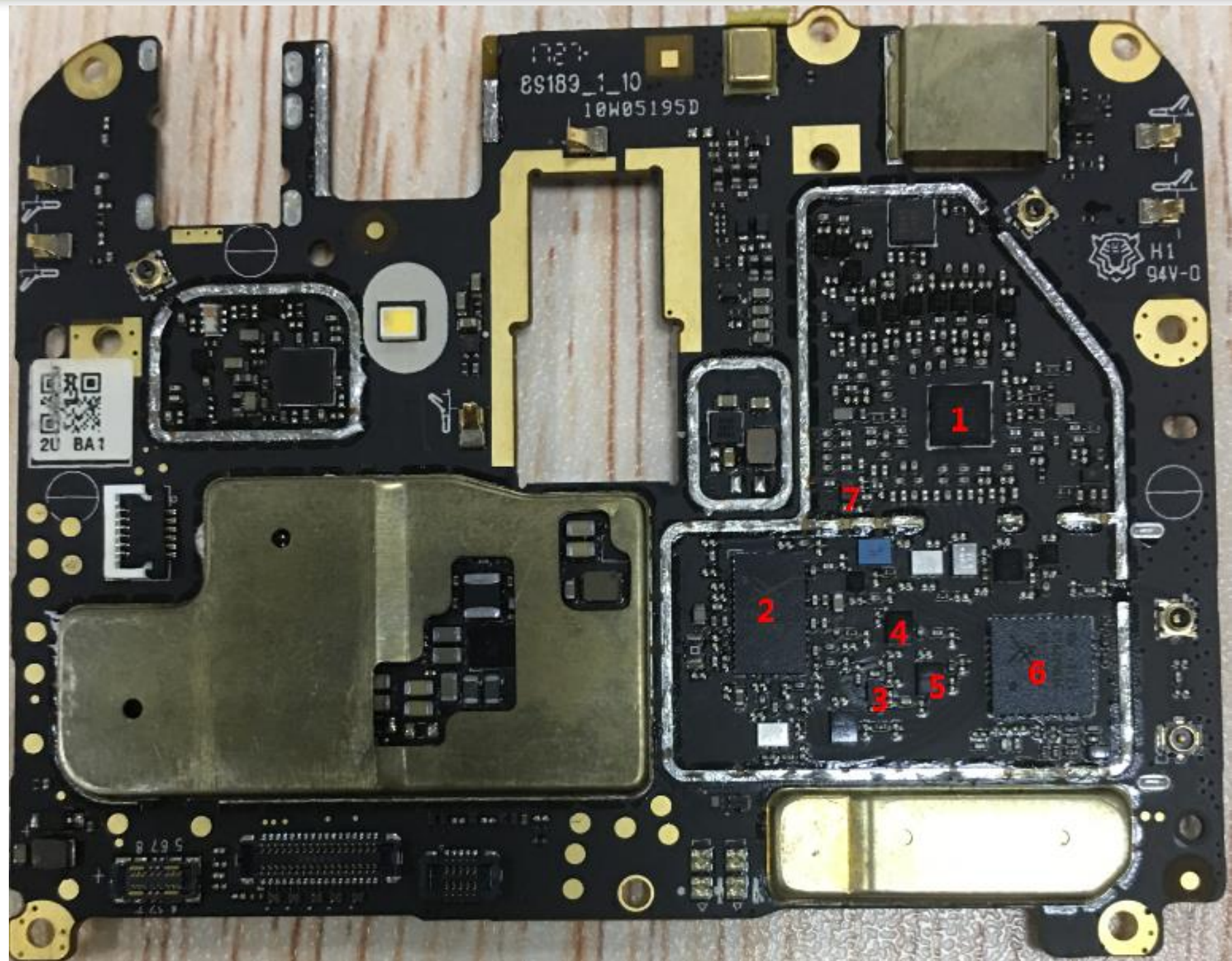
1. U2200 : Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2520 : TX Band Switch
4. U2528 : B28A Duplexer
5. U2529 : B28B Duplexer
6. U2402 : TX Module
[Sky77912-61](#)
7. U2506 : PRX Band Switch

Serial IC

Tx comp.

Rx comp.

Tx/Rx comp.



TD-LTE Band 38/41/40

Tx path

Rx path

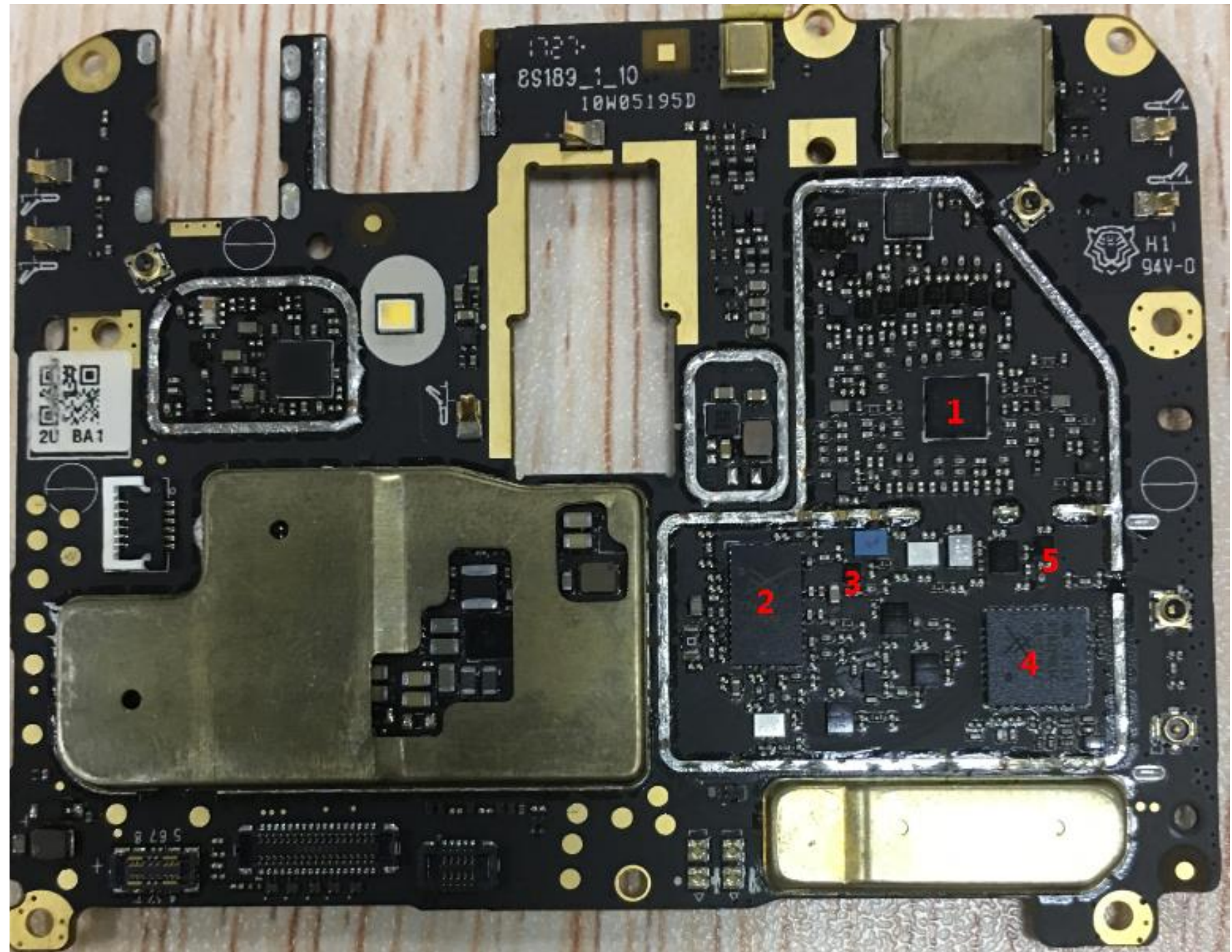
1. U2200: Transceiver
[WTR2965](#)
2. U2302 : Power Amp.
[Sky77645-11](#)
3. U2303 : B38/41/40 TX SAW
4. U2402: TX Module
[Sky77912-61](#)
5. U2540 : B38/41/40 PRX SAW

Serial IC

Tx comp.

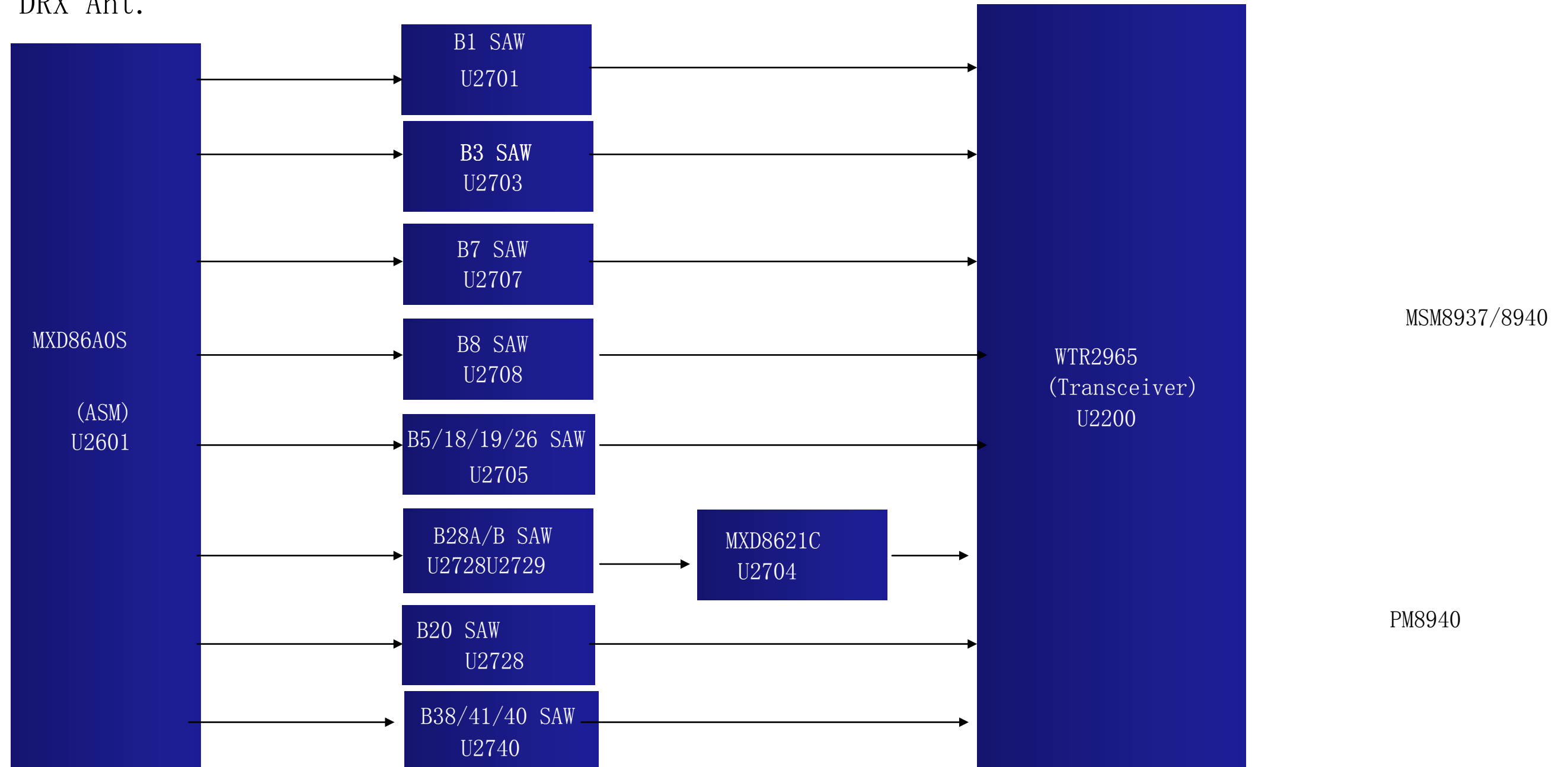
Rx comp.

Tx/Rx comp.



LTE Div. Rx Block Diagram

DRX Ant.



LTE Band 1 DRx

Tx path

Rx path

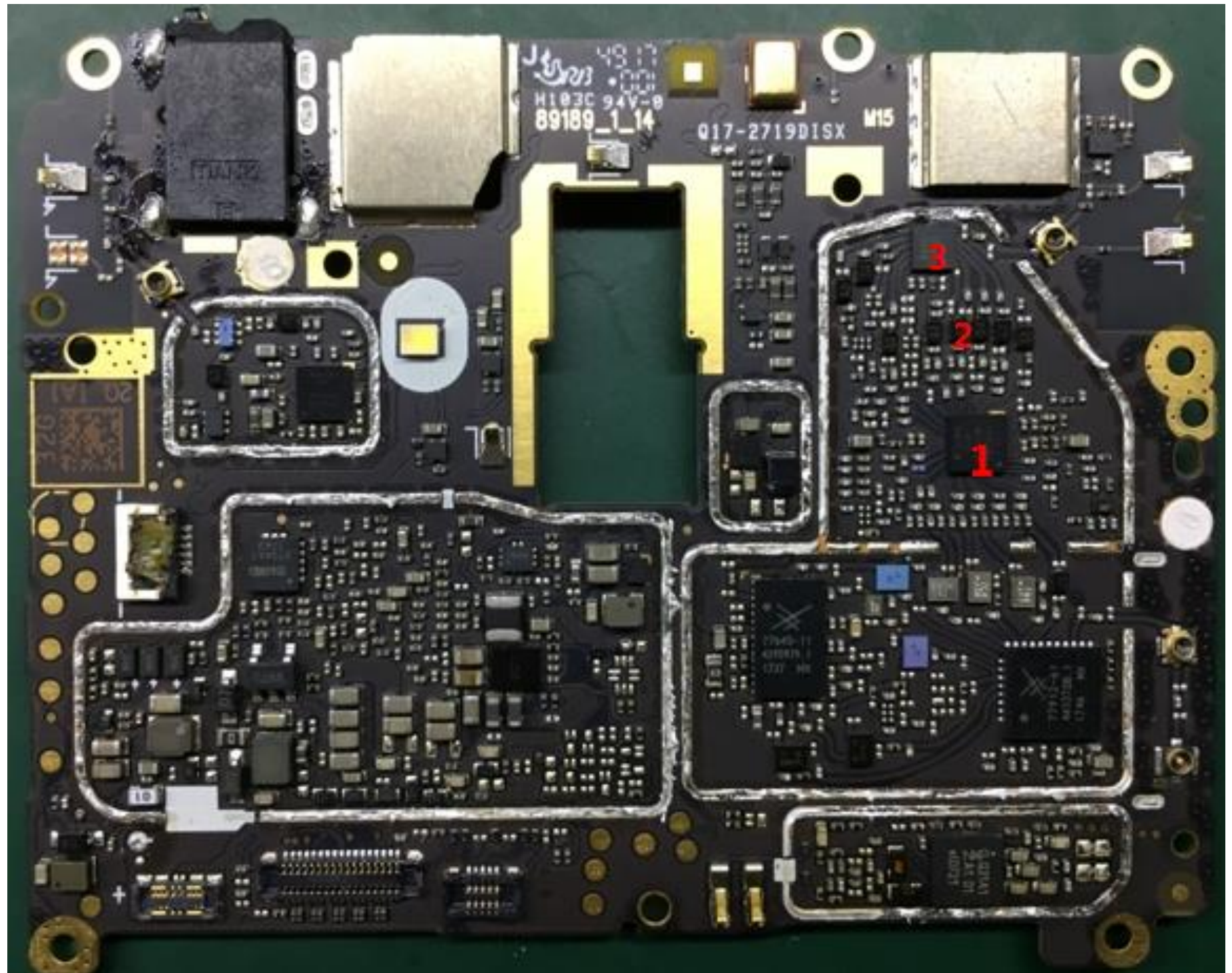
1. U2200: Transceiver
WTR2965

2. U2701 : B1 saw

3. U2601: Ant. Switch
MXD86A0S

Serial IC

Rx comp.



LTE Band 3 DRx

Tx path

Rx path

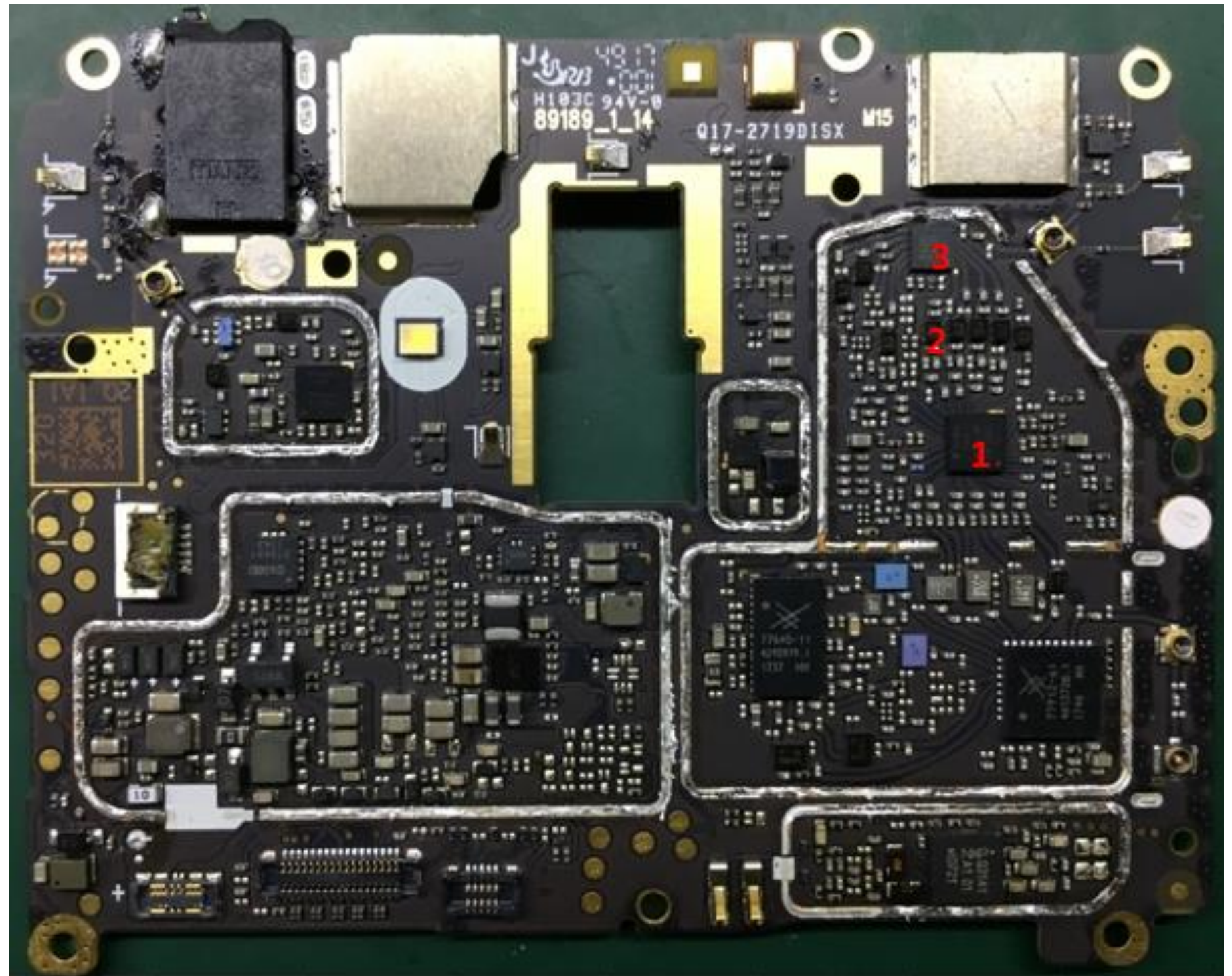
1. U2200: Transceiver
WTR2965

2. U2703 : B3 saw

3. U2601: Ant. Switch
MXD86A0S

Serial IC

Rx comp.



LTE Band 5/18/19/26 DRx

Tx path

Rx path

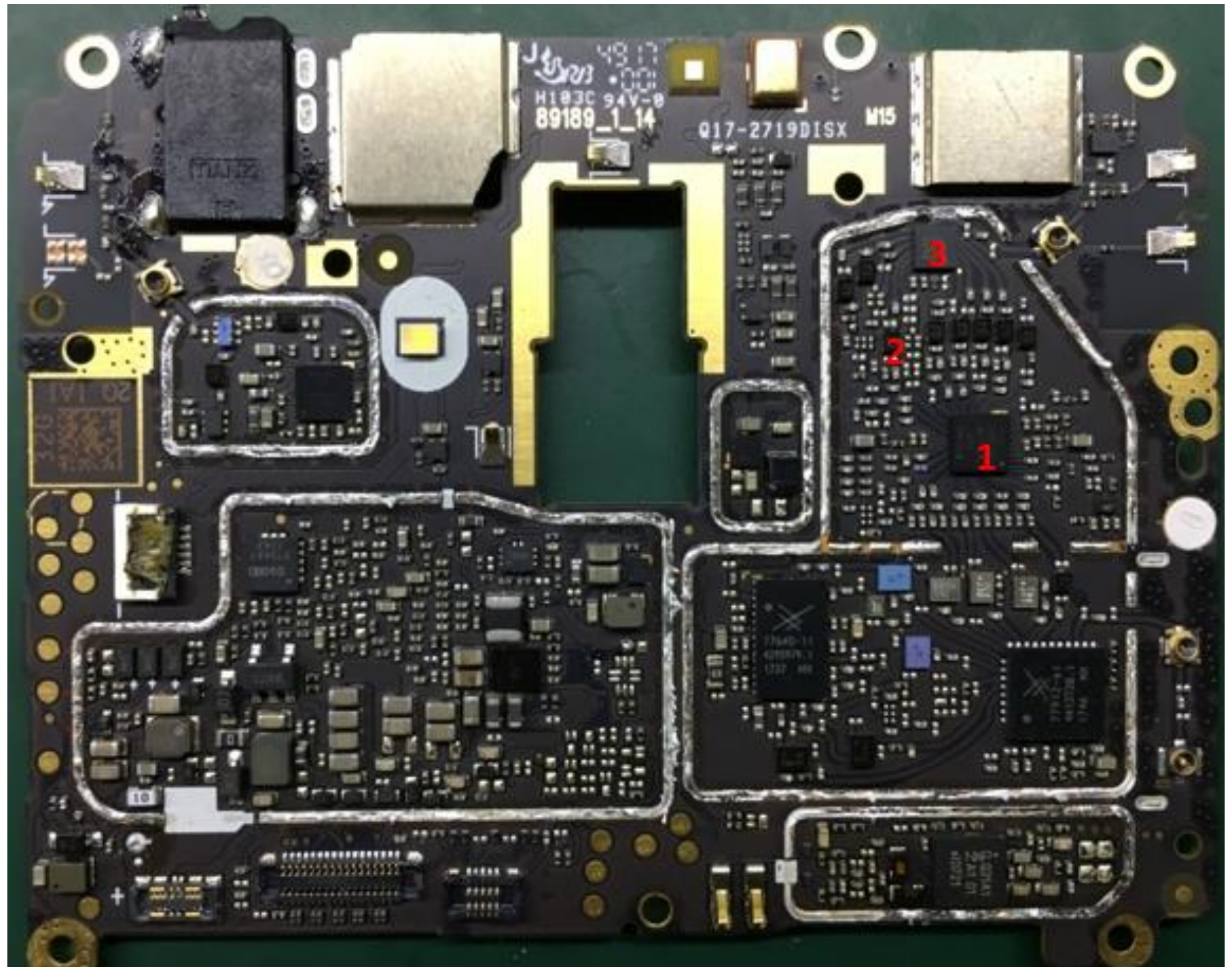
1. U2200: Transceiver
WTR2965

2. U2705 : B5 saw

3. U2601: Ant. Switch
MXD86A0S

Serial IC

Rx comp.



LTE Band 7 DRx

Tx path

Rx path

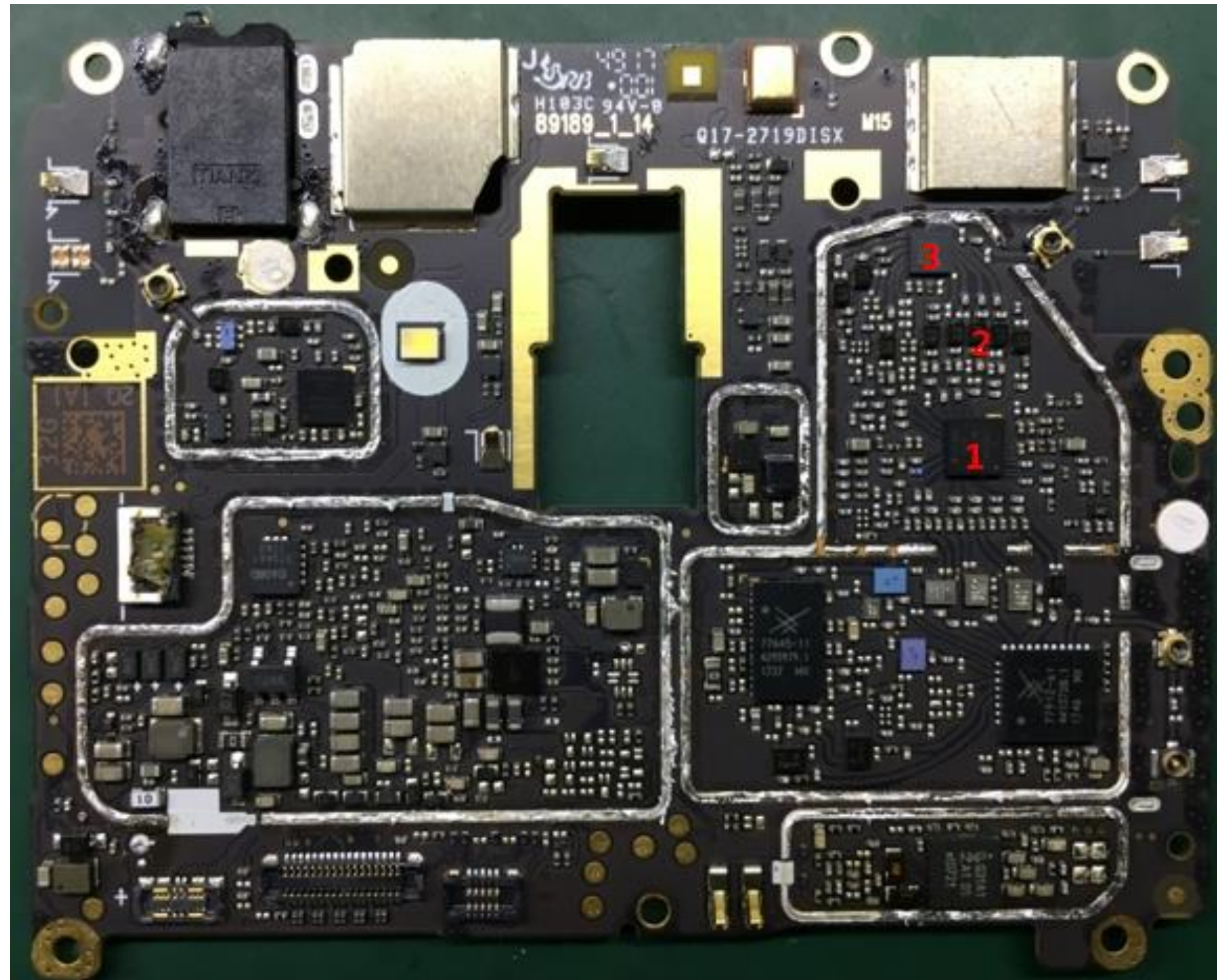
1. U2200: Transceiver
WTR2965

2. U2707 : B7saw

3. U2601: Ant. Switch
MXD86A0S

Serial IC

Rx comp.



LTE Band 8 DRx

Tx path

Rx path

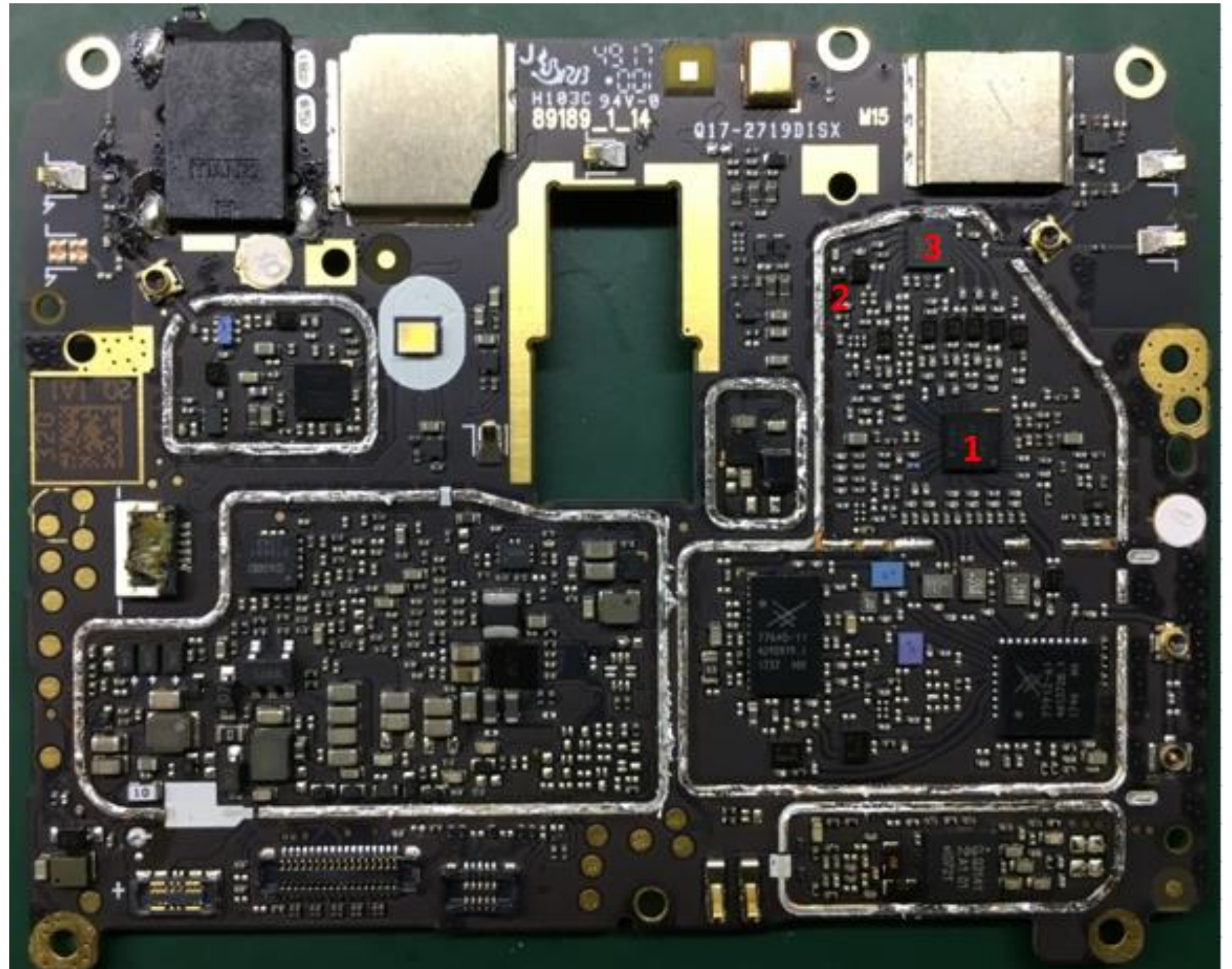
1. U2200: Transceiver
[WTR2965](#)

2. U2708 : B8saw

3. U2601: Ant. Switch
[MXD86A0S](#)

Serial IC

Rx comp.



LTE Band 20 DRx (WW)

Tx path

Rx path

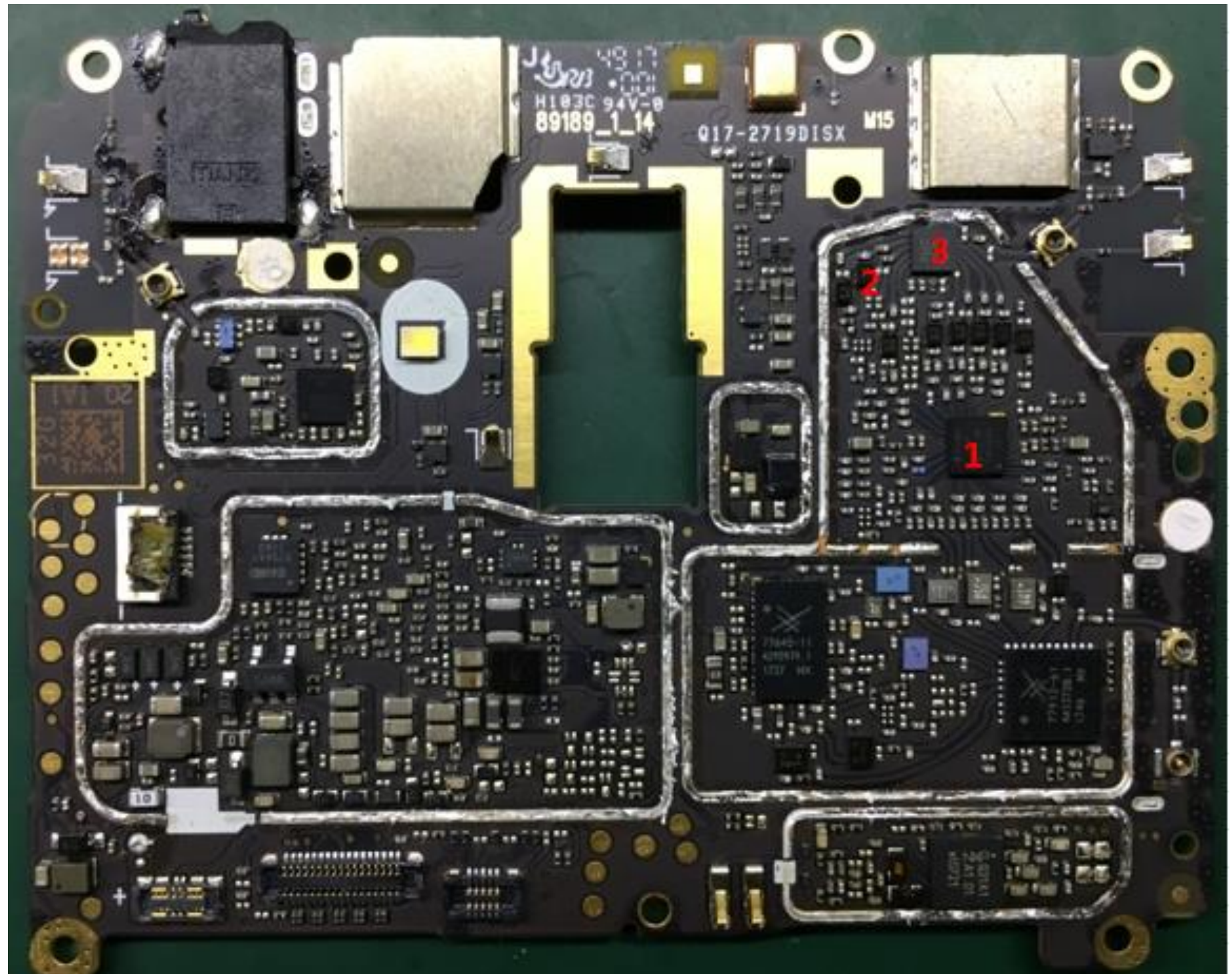
1. U2200: Transceiver
WTR2965

2. U2728 : B20saw

3. U2601: Ant. Switch
MXD86A0S

Serial IC

Rx comp.



LTE Band 28 DRx (TW)

Tx path

Rx path

1. U2200: Transceiver

WTR2965

2. U2704 : Band Switch

3. U2728 : B28A saw

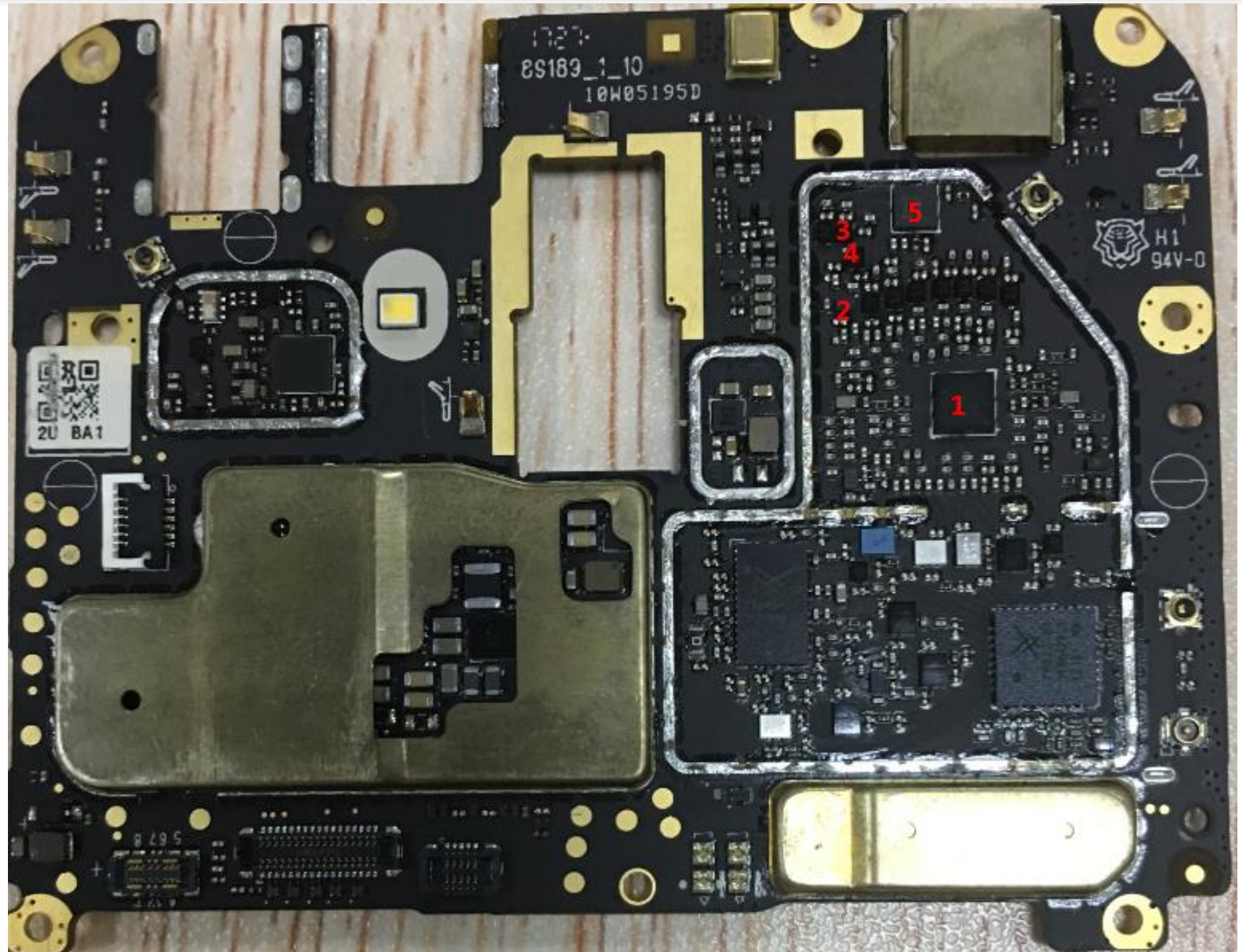
4. U2729 : B28B saw

5. U2601 : Ant. Switch

MXD86A0S

Serial IC

Rx comp.



LTE Band 38/41/40 DRx

Tx path

Rx path

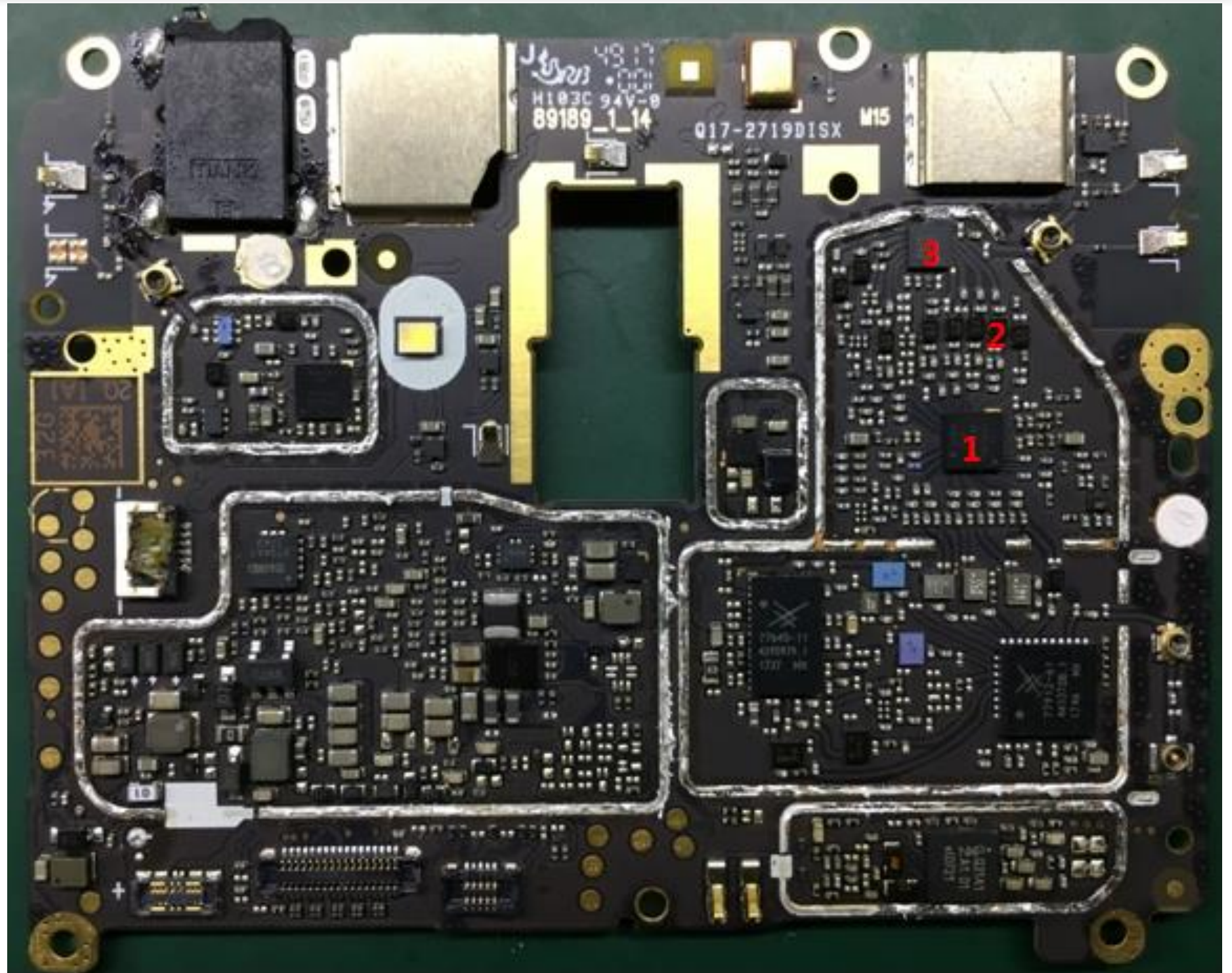
1. U2200: Transceiver
[WTR2965](#)

2. U2740 : B38/41/40saw

3. U2601: Ant. Switch
[MXD86A0S](#)

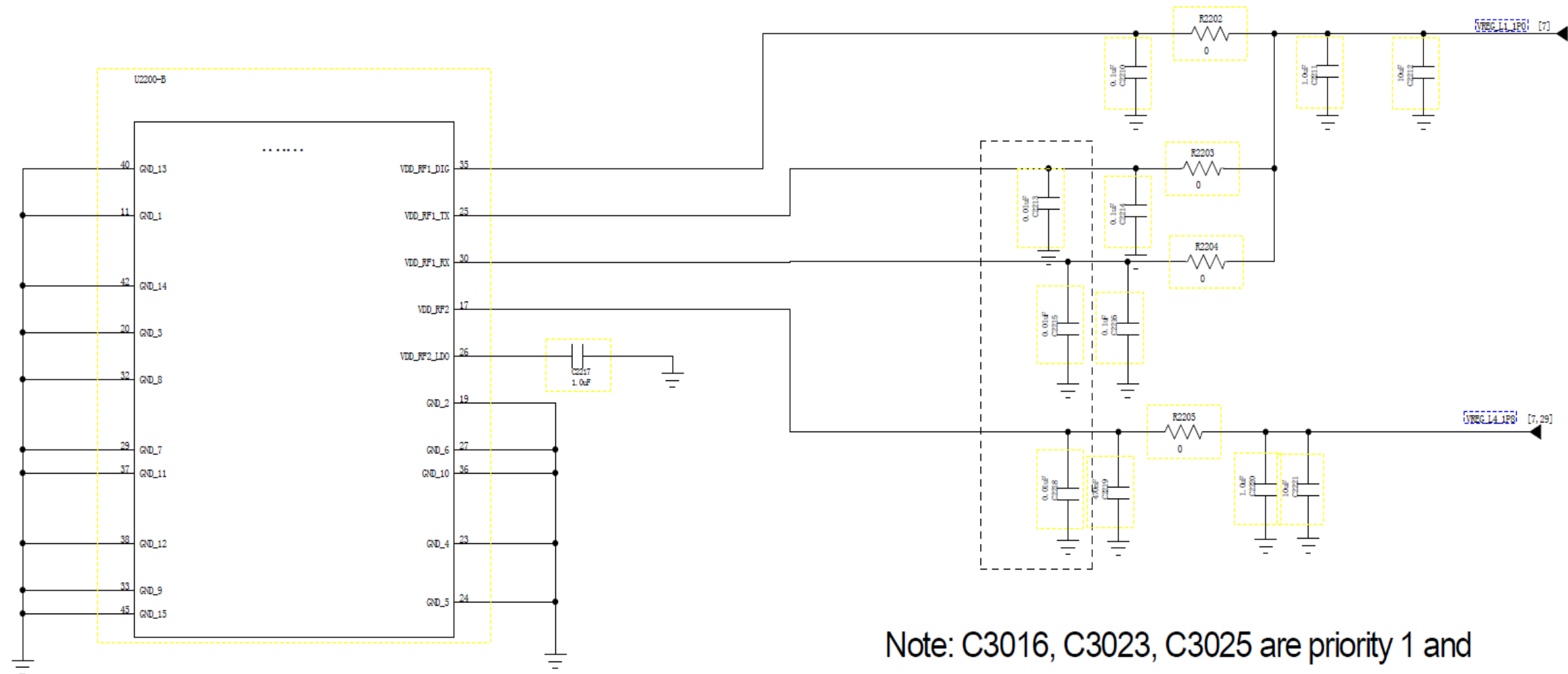
Serial IC

Rx comp.



Voltage Setup

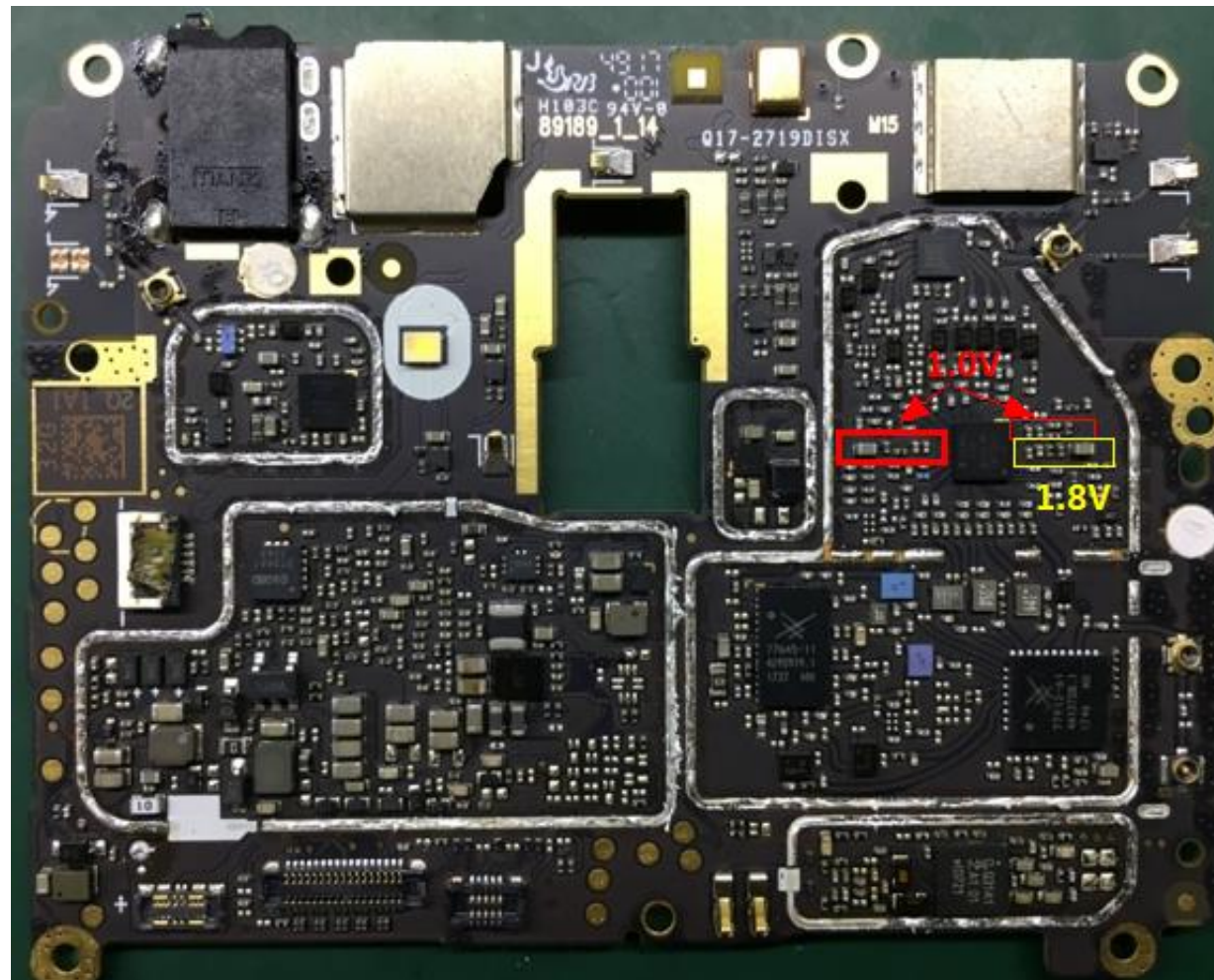
Transceiver



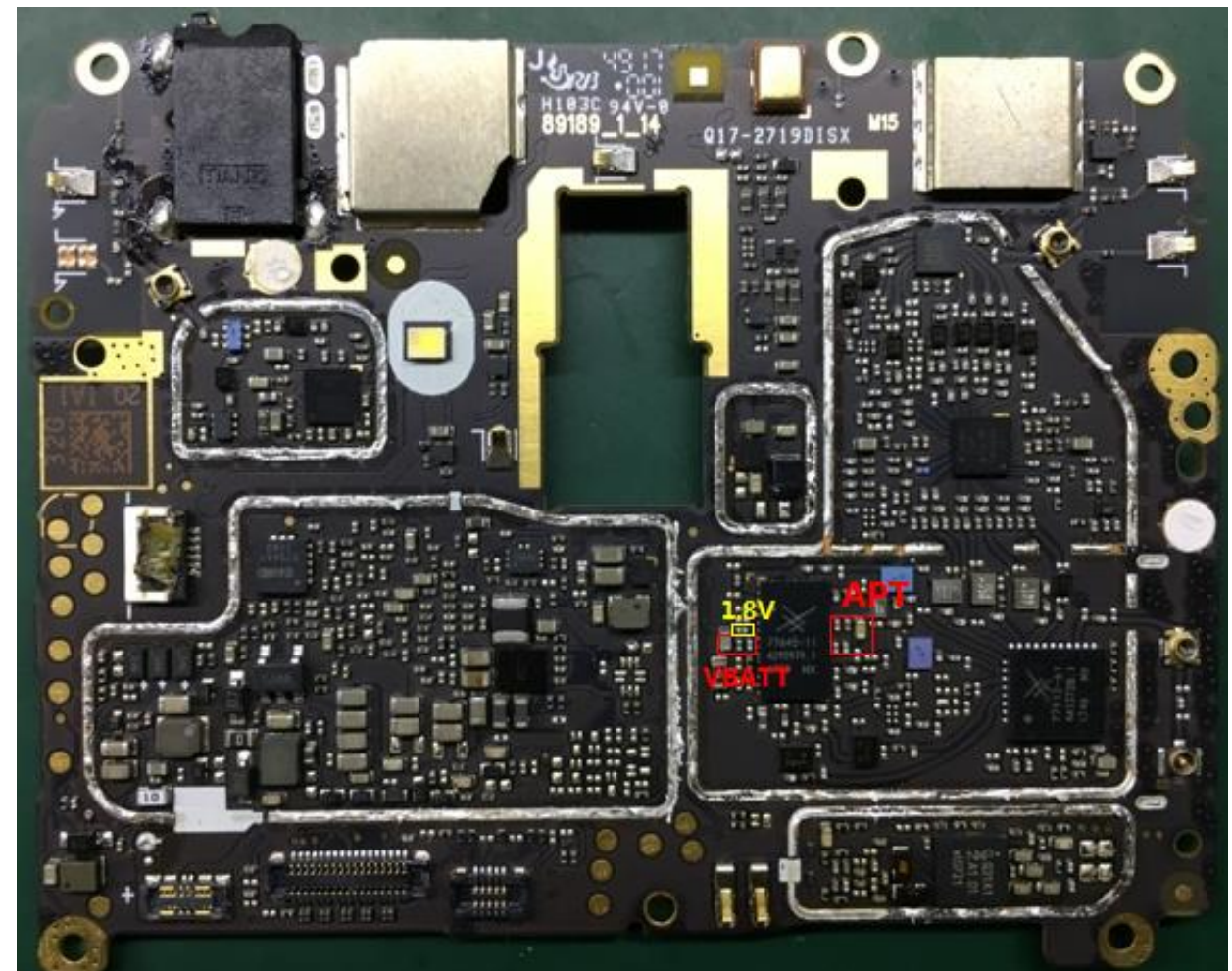
Note: C3016, C3023, C3025 are priority 1 and

Voltage Setup

Transceiver



3G/4G PA



Outline

(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/Band8/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

2G RF Test

Error CodeAction

Tx Power

1. Re Calibration
2. Check Tx Path
3. Change TXM (U2402)

ORFS/
Power vs. Time

1. Check/change test station fixture
2. Check/change calibration station fixture
3. Check Tx Path
4. Change TXM (U2402)

Frequency Error/
Phase Error

1. Check/change PMIC

Sensitivity1. Check/Change PMIC

2. Check/Change crystal

3G RF Test

Error CodeAction

Tx Power/ACLR/ SEM/OBW/EVM	1. Re calibration 2. Check Tx Path 3. Change PA (U2302)
Sensitivity	1. Re calibration 2. Check Rx Path
Frequency Error	1. Check/Change PMIC 2. Check/Change crystal

4G RF Test

Error CodeAction

Tx Power/ACLR/ SEM/OBW/EVM	1. Re calibration 2. Check Tx Path 3. Change PA (U2302)
Sensitivity	1. Re calibration 2. Check Rx Path
Frequency Error	1. Check/Change PMIC 2. Check/Change crystal

以下零件若有動到，需重新做Calibration(PHONE)

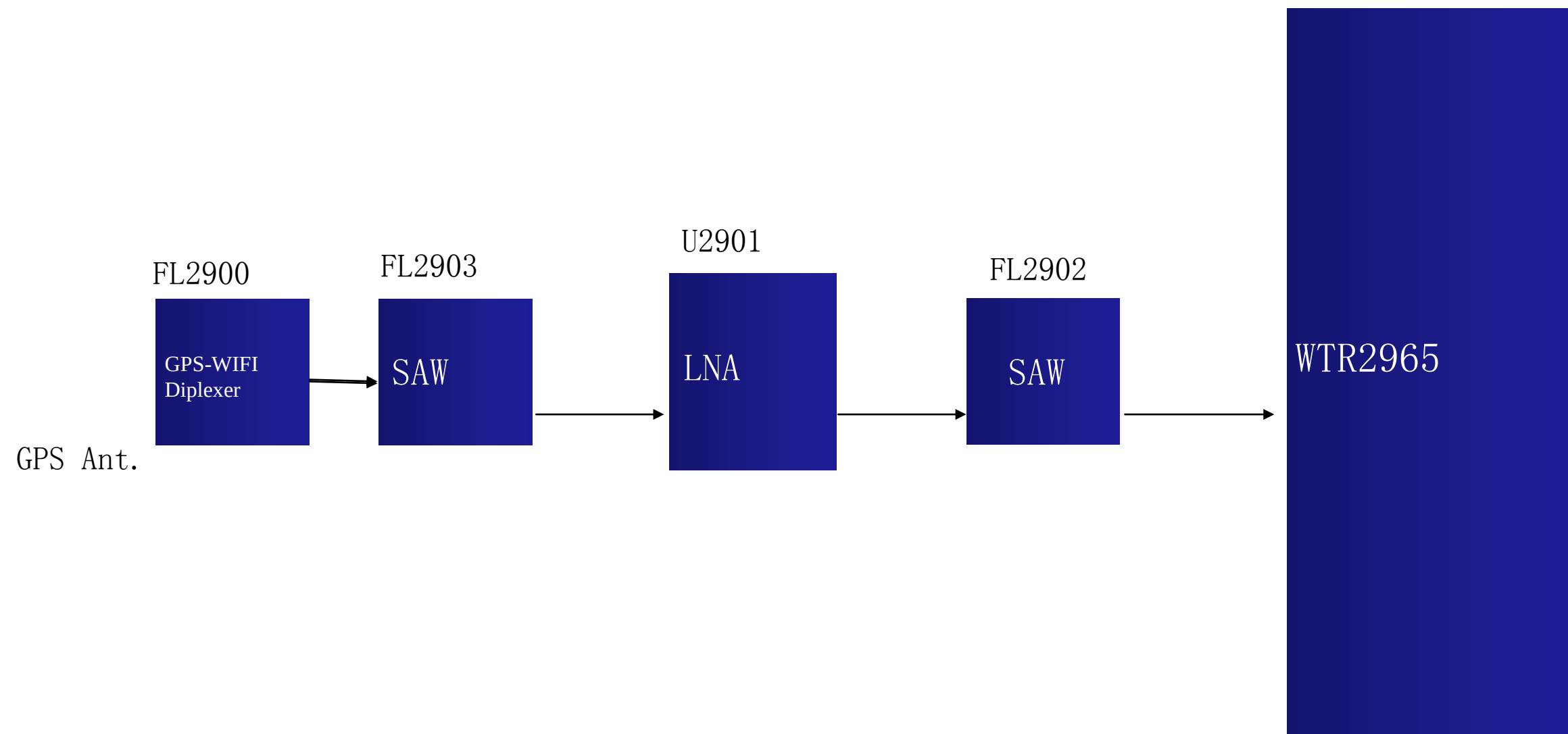
1. Transceiver: U2200; TXM: U2402; PMIC: U0401
2. PA: U2302
3. Duplexer: U2501 U2502 U2503 U2507 U2508 U2528 U2529
4. SAW: U2303 U2540 U2506 U2701 U2702 U2703 U2705
U2708 U2728 U2729 U2704 U2740

Outline

(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/Band8/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

GPS Ant Block Diagram



GPS

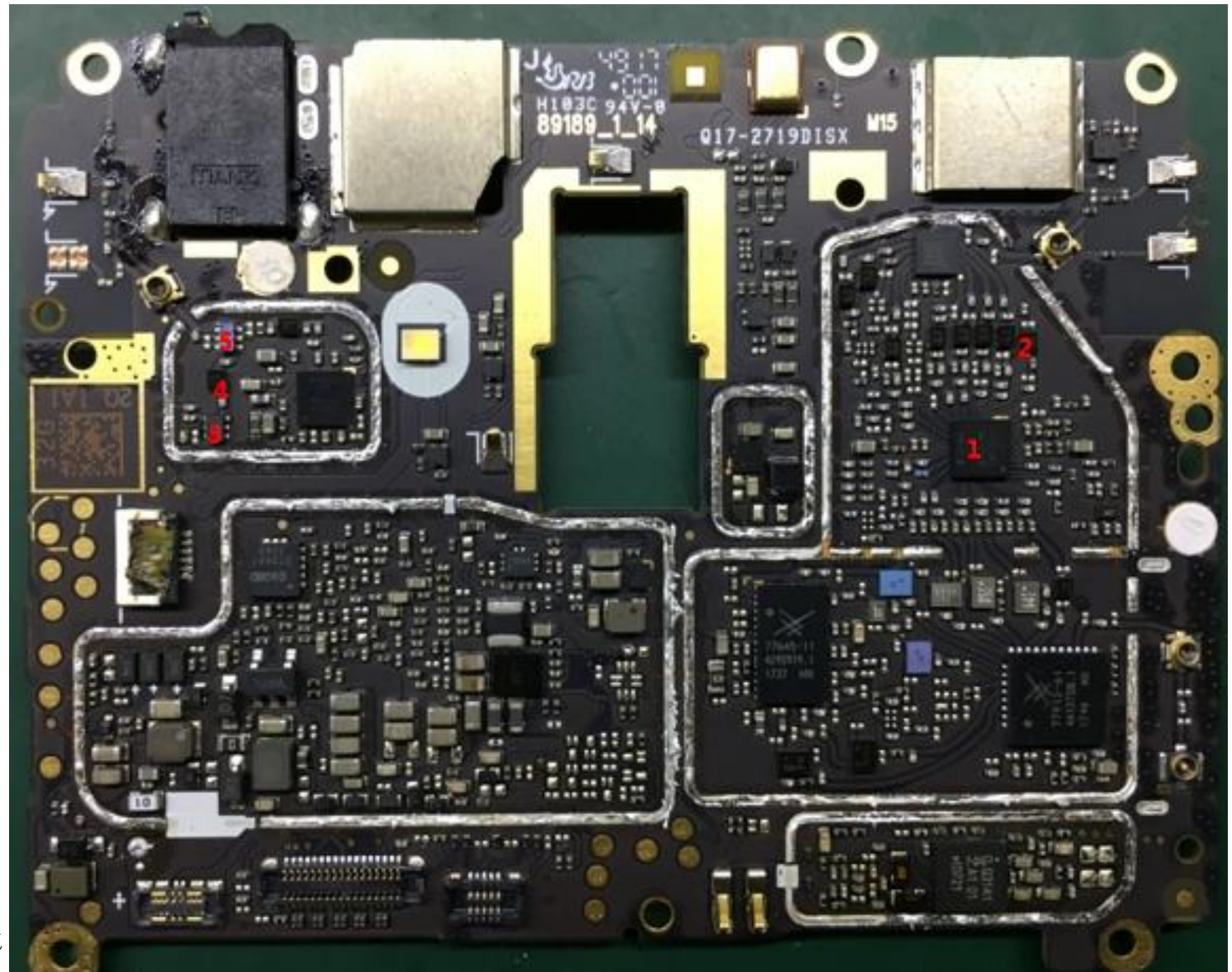
1. U2200: Transceiver
WTR2965
2. FL2902: GPS SAW Filter
3. U2901: GPS LNA
AW5005DNR
4. FL2903: GPS SAW Filter
5. FL2900: GPS-WIFI Diplexer

GPS Antenna area

Serial comp.

with orientation

Serial comp. without
orientation



GPS Function Test Procedure

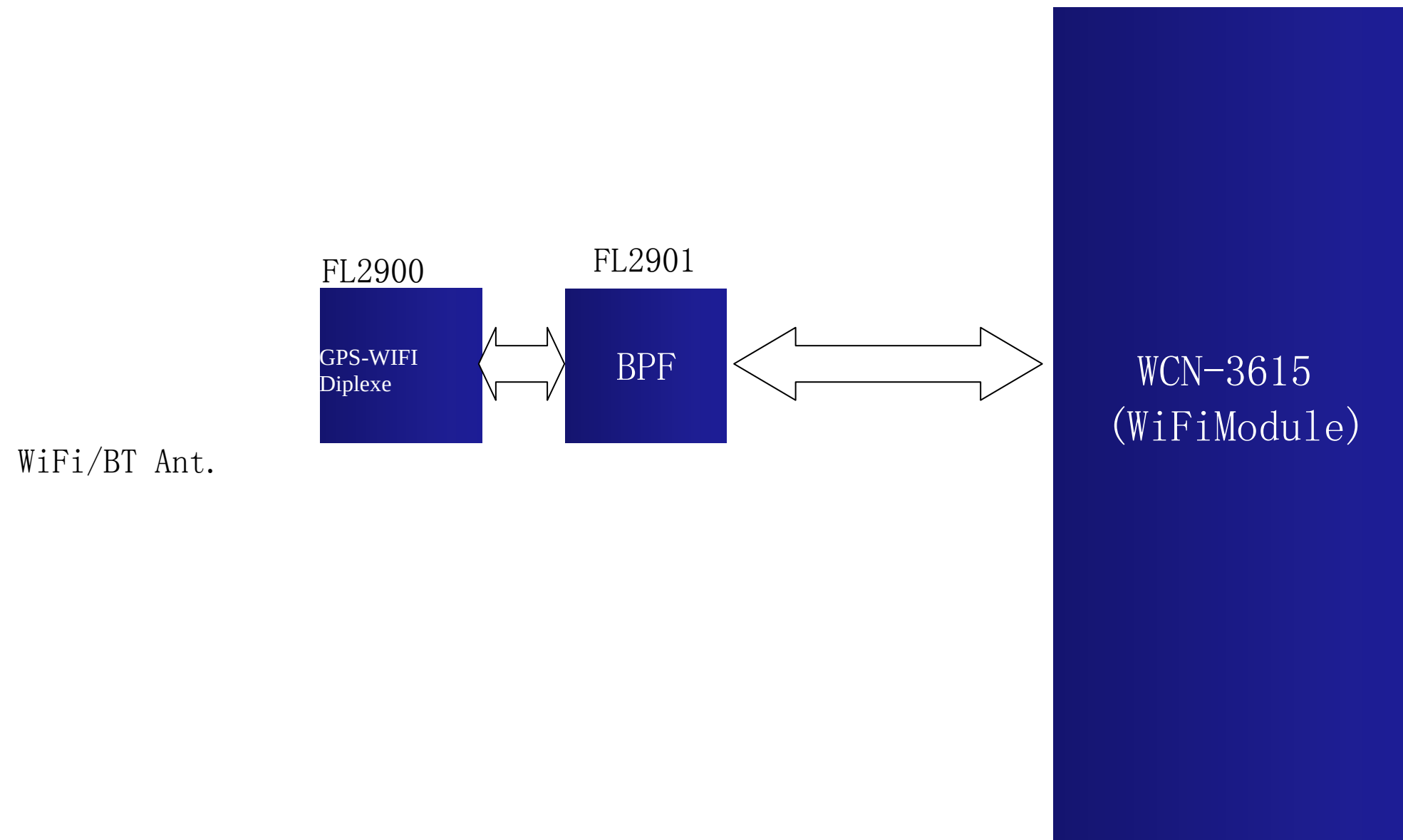
- Check GPS function with GPS Test Tool.
- Check the GPS feed point.
- Check U2901 and surrounding components
- Check the 1.8V supply voltage of GPS LNA

Outline

(*) Debug Tool Required

- Repair and Test Environment (*)
- RF Block Diagram
- RF Calibration Repair Guide (*)
 - GSM Quad-bands Block Diagram
 - GSM EGSM and GSM850
 - GSM DCS and PCS
 - WCDMA Quad-bands Block Diagram
 - WCDMA Band1/Band2/Band5/Band6/Band8/Band19
 - LTE Quad-bands Block Diagram
 - LTE Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
 - LTE Div. Rx Block Diagram
 - LTE Div. Rx Band 1/3/5/7/8/18/19/20/26/28 & TD-LTE 38/40/41
- RF Test Trouble Shooting
 - 2G RF Test
 - 3G RF Test
 - 4G RF Test
- GPS Repair Guide
- BT/WiFiRepair Guide

WiFi/BT Ant Block Diagram



WiFi/BT

WiFi Antenna area

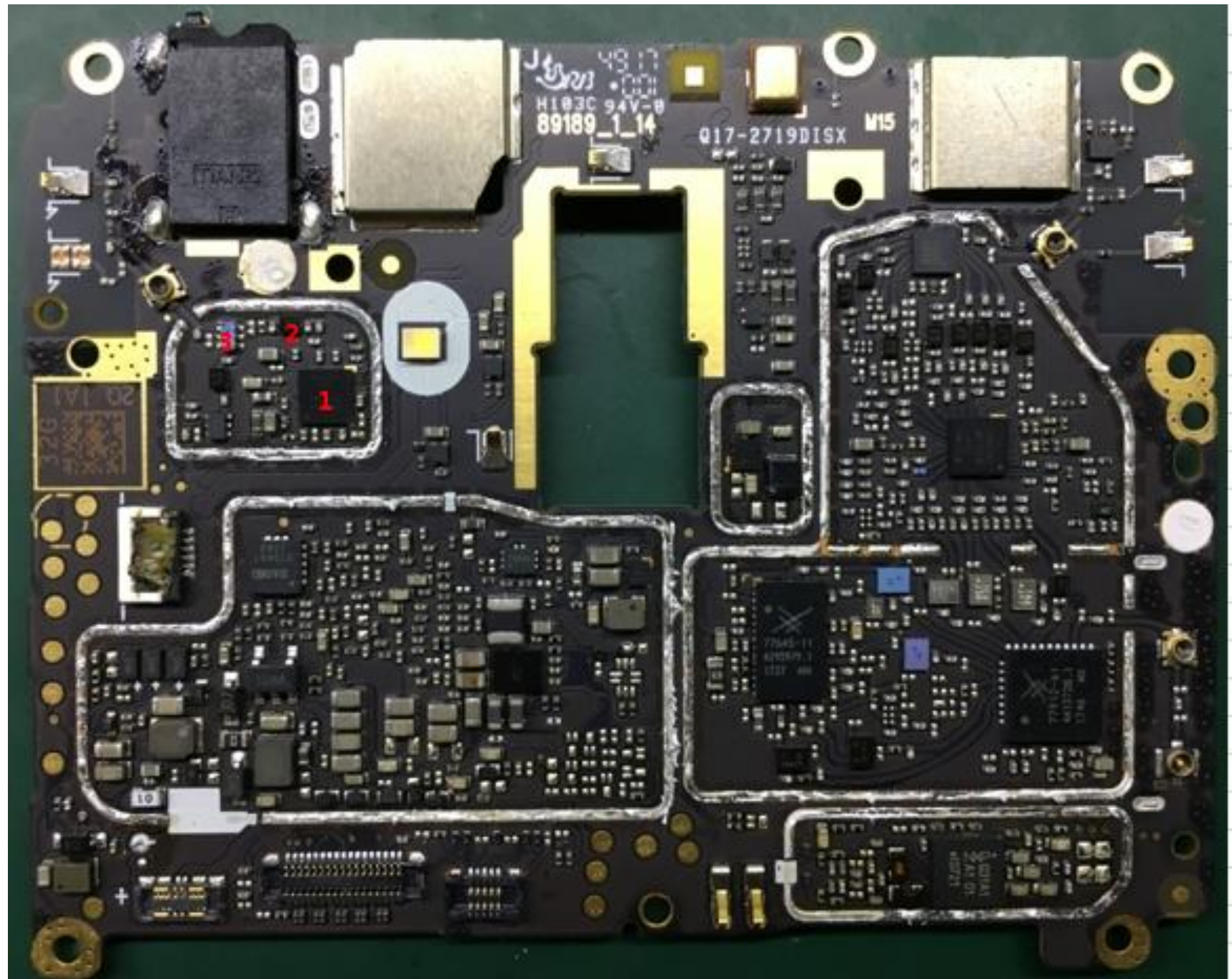
Serial comp.
without orientation

Serial comp.
with orientation

1. U2900: WCN3615

2. FL2901: WIFI/BT SAW

3. FL2900: WIFI/BT/GPS Diplexer



BT/WLAN Function Test Procedure

- Change antenna
- Check the BT/WLAN feed point
- Check WCN3615 and surrounding components

Thanks for Your attentions !

ASUS[®] Inspiring Innovation • Persistent Perfection

Inspiring Innovation. Persistent Perfection